Bridging Lexical and Semantic Retrieval for Legal Research Support: A Retrieval-Augmented Generation Approach to Question Answering over Indian Supreme Court Precedent

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Abstract

Within the Indian judiciary, where Article 141 renders Supreme Court judgments binding precedent nationwide and case backlogs are substantial, legal research still depends heavily on keyword- and Boolean-based search, which performs poorly when a query and a relevant precedent use different vocabulary for the same concept. Controlled comparisons linking retrieval strategy to downstream generation quality and abstention behavior have been lacking in this setting. NyayaRAG was developed to close this gap, comparing six retrieval configurations — TF-IDF, BM25, three dense embedding models (E5, BGE, InLegalBERT), and a Hybrid BM25–E5 retriever fused via Reciprocal Rank Fusion — within one retrieval-augmented generation pipeline, and evaluating the system's ability to generate evidence-grounded answers and abstain when evidence was insufficient. The corpus comprised 782 English-language 2024 Supreme Court judgments segmented into 30,521 chunks via a custom paragraph-aware chunker. Retrieval was evaluated against a 135-question benchmark from 29 judicial officers; generation and abstention were evaluated on an independent 60 Answerable/60 Unanswerable benchmark using Google's Gemini (gemini-3.6-flash) for citation-constrained generation. Semantic retrieval did not significantly outperform lexical retrieval (Wilcoxon signed-rank, three metrics, all non-significant, negligible effects), but Hybrid BM25–E5 significantly outperformed every individual retriever (Friedman chi-square, p ¡ .001; Holm-corrected comparisons, predominantly large effects) and was carried forward for generation. Answers reached a mean composite quality of 0.8983, significantly exceeding a 0.75 benchmark (Wilcoxon W

= 1452, p = .000013, r = 0.59), while the system safely abstained on 98.33 percent of evidence-insufficient questions, exceeding an 80 percent benchmark (exact binomial p = .000025, Cohen's h = 0.67). These findings support NyayaRAG as a feasible, citation-transparent, abstention-capable research-support tool for judicial officers, advocates, and legal researchers working with 2024 Indian Supreme Court case law.

Keywords: retrieval-augmented generation, legal question answering, Indian Supreme Court judgments, hybrid retrieval, dense passage retrieval, abstention, citation-grounded generation

1 Introduction

1.1 Background and Context

India's judiciary operates under considerable structural strain: more than 23,000 judges across all tiers of the system confront a longstanding, well-documented backlog of pending cases (Nigam et al., 2023). This strain is compounded by Article 141 of the Constitution of India, under which judgments issued by the Supreme Court constitute binding law for every court in the country. As a direct consequence, the ability to retrieve relevant precedent quickly and accurately is not simply a matter of convenience for legal professionals, but a functional requirement of the judicial system itself.

Despite this requirement, legal research in India has historically relied on keyword- and Boolean-based search over judgment databases. Such methods match surface-level tokens rather than underlying legal meaning, and consequently perform poorly whenever a query and a relevant judgment express the same concept using different vocabulary (Martinez-Gil, 2023). Legal natural language processing has since progressed toward Transformer-based representation learning and, more recently, retrieval-augmented generation (RAG) architectures that pair a retrieval component with a generative model (Hartung et al., 2023). Conditioning a generative model on retrieved evidence, however, does not by itself guarantee correctness: a model supplied with irrelevant or misapplied evidence can still produce a fluent but legally incorrect response. Benchmarking of deployed commercial legal-AI tools indicates that this risk is not merely theoretical, since such systems have been shown to hallucinate case citations despite marketing claims of grounded output (Magesh et al., 2024), and recent surveys of the field argue that reliability and explainability — rather than fluency — should determine whether a legal AI system can be trusted (Nguyen et al., 2025).

1.2 Problem Statement

Retrieval-augmented generation, as formalized by Lewis et al. (2020), addresses this problem in principle by grounding a generative model's output in evidence retrieved from an external corpus rather than in parametric knowledge alone, thereby reducing factual drift. The retrieval component of such a pipeline can draw on several distinct

paradigms. Lexical ranking functions such as BM25 remain a strong, computationally inexpensive baseline (Robertson Zaragoza, 2009), while dense dual-encoder retrieval has been shown to outperform lexical methods where query and passage vocabulary diverge (Karpukhin et al., 2020). Dense embedding design has since diversified into contrastively pre-trained, general-purpose models such as E5 (Wang et al., 2022) and BGE (Xiao et al., 2024), as well as domain-adaptive approaches such as InLegalBERT, which is pre-trained directly on Indian legal text (Paul et al., 2023). A further strategy, hybrid retrieval, fuses lexical and dense rankings — for example via Reciprocal Rank Fusion — to combine exact-term precision with semantic recall; evidence from heterogeneous benchmarks such as BEIR indicates that no single retrieval paradigm dominates uniformly across domains (Thakur et al., 2021). Beyond retrieval, the literature further establishes that retrieval quality substantially constrains the answer quality achievable by a legal RAG system regardless of the generation model used (Pipitone Houir Alam, 2024), that prompt structure materially affects task performance in legal RAG settings (Sriram et al., 2026; Trautmann et al., 2022; Yu et al., 2022), that chunk boundaries are a first-order determinant of downstream answer reliability rather than an incidental implementation detail (Bennani Moslonka, 2026; Bhat et al., 2025), and that a system's capacity to recognize insufficient evidence and decline to answer is itself a core reliability property (Wen et al., 2024).

Taken together, this body of work establishes that semantic and hybrid retrieval generally outperform purely lexical retrieval in adjacent settings (Nigam et al., 2023; Karpukhin et al., 2020), that retrieval performance constrains what a downstream generation stage can achieve, and that hallucination remains a risk even within retrieval-grounded systems, making abstention a necessary rather than optional design property. What this literature does not provide is a controlled comparison of multiple lexical, dense, and hybrid retrieval configurations conducted on a single expert-constructed Indian Supreme Court benchmark, nor an account that statistically connects retrieval-stage performance to both generation-stage answer quality and abstention reliability within one evaluated system. More specifically, prior research has not (a) compared lexical, dense, and hybrid retrieval within a single experimental design built on a shared corpus, chunking scheme, and benchmark; (b) relied on a benchmark whose questions and relevance judgments were contributed directly by practicing judicial officers rather than assembled from crowd-sourced or synthetic queries; or (c) statistically linked retrieval-stage outcomes to both generation quality and abstention behavior under matched conditions.

1.3 Purpose of the Study

This study addresses that gap by designing, implementing, and empirically evaluating NyayaRAG, a retrieval-augmented generation framework for legal question answering over Indian Supreme Court judgments. Its purpose is to determine which retrieval strategies most effectively surface relevant precedent and to establish how retrieval quality translates into evidence-grounded, trustworthy generated answers, using statistically validated evidence rather than descriptive comparison. To isolate retrieval strategy as the sole manipulated variable, all other experimental factors — the underlying corpus, the chunking scheme, the benchmark, a fixed retrieval depth of $k = 5$,

and a generation temperature of 0.0 — are held constant across the configurations compared. A further aim is to assess whether the resulting system can generate accurate, evidence-grounded answers while reliably abstaining when available evidence is insufficient. In pursuing these aims, the study is intended to yield both a reusable evaluation methodology for legal RAG systems more broadly and a working, benchmarked prototype relevant to the Indian judicial context.

1.4 Research Questions

The study is guided by the following research questions:

- **RQ1:**Do semantic retrieval models significantly outperform lexical retrieval models for Indian Supreme Court legal question answering?
- **RQ2:**Which single retrieval model performs best overall among TF-IDF, BM25, E5, BGE, InLegalBERT, and a Hybrid BM25–E5 configuration?
- **RQ3:**How accurately does NyayaRAG generate evidence-grounded answers?
- **RQ4:**Can NyayaRAG reliably detect insufficient evidence and abstain from answering?

1.5 Contributions and Expected Value

The study's contributions can be distinguished at several levels. Methodologically, it contributes a statistically rigorous evaluation protocol — combining Wilcoxon signed-rank tests, Friedman tests with Holm-corrected post-hoc comparisons, and exact binomial tests — that connects retrieval-stage performance to generation-stage answer quality and abstention behavior, offered as a reusable approach for evaluating legal RAG systems generally. At the technical and system level, the study contributes a controlled, six-configuration comparison of retrieval strategies (TF-IDF, BM25, E5, BGE, InLegalBERT, and a Hybrid BM25–E5 retriever combined via Reciprocal Rank Fusion) implemented within a single RAG pipeline, together with the NyayaRAG system itself, which retrieves relevant precedent, generates answers that cite their supporting evidence, and is designed to recognize when available evidence does not support a confident answer and to abstain rather than produce an unsupported response; traceability of every generated answer to specific retrieved text is treated as a core design requirement, directly addressing trustworthiness concerns raised in the legal-AI literature (Nguyen et al., 2025; Magesh et al., 2024). Empirically, the study contributes an expert-generated benchmark constructed from questions and relevance judgments supplied by practicing judicial officers rather than crowd-sourced or synthetic queries. Practically, the expected value of this work accrues primarily to judicial officers, advocates, and legal researchers, the professional user group around which the benchmark was built; the study does not evaluate the system with citizen or lay-user participants, and no claim is made regarding its suitability for direct citizen-facing legal advice, an application that would require separate, dedicated evaluation.

2 Literature Review

2.1 Literature Review Approach

The review synthesizes material drawn from ACL Anthology, arXiv, IEEE Xplore, and Google Scholar, complemented by focused searches of venues specializing in legal technology and information retrieval. Search terms combined legal-domain vocabulary (e.g., "legal NLP," "Indian judiciary," "case law retrieval," "Article 141") with technically oriented terms (e.g., "retrieval-augmented generation," "dense passage retrieval," "hallucination," "faithfulness," "abstention"). Of the 27 sources examined, 10 were judged directly foundational to the study's research questions and technical design and are consolidated in Table 1; the remaining sources contributed background framing and are referenced narratively throughout this review and in the Materials and Method section.

2.2 Foundational Research

Several distinct research streams converge to motivate and technically ground the present study.

2.2.1 Retrieval-augmented generation as a factuality mechanism.

Lewis et al. (2020) addressed the tendency of purely parametric language models to produce unsupported output by conditioning generation on retrieved evidence, combining dense retrieval with a BART-based generator. Their central finding — that retrieval-augmented generation improves factual grounding relative to generation from parametric knowledge alone — supplies the conceptual foundation for the present study's third and fourth research questions, which examine the accuracy and grounding of generated answers and the reliability of abstention when evidence is lacking.

2.2.2 Dense retrieval as an alternative to lexical matching.

Karpukhin et al. (2020) introduced a dual-encoder dense passage retrieval (DPR) architecture and demonstrated that it outperforms lexical retrieval methods for open-domain question answering, particularly where query and passage vocabulary diverge. This result directly informs the design of the dense retriever component evaluated under the present study's first research question.

2.2.3 Lexical retrieval as a durable baseline.

Robertson and Zaragoza (2009) established the BM25 ranking function as a strong and computationally inexpensive lexical baseline for probabilistic information retrieval. BM25 accordingly serves as the lexical baseline against which semantic retrieval methods are compared in the present study's first and second research questions.

2.2.4 Diversification of embedding-model design.

Two further methodological streams elaborate on dense representation learning. Wang et al. (2022) proposed E5, a family of embeddings trained via large-scale contrastive

pre-training on general-domain text pairs, and reported improved semantic retrieval as a result. Xiao et al. (2024) introduced BGE, a packaged general-purpose embedding resource shown to perform strongly across multiple domains. Both models inform the dense retriever configurations evaluated under the present study's first and second research questions.

2.2.5 Domain-adaptive pre-training for legal text.

Paul et al. (2023) developed InLegalBERT through domain-adaptive pre-training of a BERT-family encoder directly on Indian legal text, reporting improved representation of Indian legal terminology relative to general-purpose encoders. This model constitutes a third dense retriever configuration evaluated in the present study.

2.2.6 Indian legal question answering.

Nigam et al. (2023) compared lexical and semantic retrieval combined with large language models in an Indian legal question-answering setting, finding that semantic retrieval outperformed lexical retrieval while also establishing that expert evaluation of generated answers — rather than retrieval metrics in isolation — was necessary to determine real-world usefulness. This finding motivates both the semantic-versus-lexical comparison underlying the present study's first research question and the expert-based answer-quality rubric used to address its third research question.

2.2.7 Benchmark design for legal RAG evaluation.

Pipitone and Houir Alam (2024) introduced the LegalBench-RAG benchmark and found that retrieval quality substantially constrains achievable legal answer quality independent of the generation model's capability. This finding directly motivated the present study's decision to treat retrieval strategy, rather than generation-model choice, as the principal experimental variable.

2.2.8 Automated evaluation of generation quality.

Es et al. (2024) proposed RAGAs, a framework demonstrating that faithfulness and relevance in retrieval-augmented generation output can be evaluated systematically through automated methods. This work informs the generation-stage evaluation methodology adopted for the present study's third research question.

2.2.9 Persistence of hallucination in commercial legal AI.

Magesh et al. (2024) benchmarked commercial legal AI tools using expert evaluation and found that such systems continue to hallucinate case citations despite marketing claims of grounded output. This finding establishes the practical need for a legal AI system capable of reliably recognizing insufficient evidence, motivating the present study's fourth research question.

Table 1 summarizes the relevance of each foundational source to the study's research questions.

Table 1 Summary of Related Work and Its Relevance to the Research Questions

Author/Year	Domain/Context	Method(s)	Key Findings	Support for RQ(s)
Lewis et al. (2020)	Retrieval-Augmented Generation	Dense retrieval + BART generation	RAG improves factuality vs. parametric-only generation	Conceptual foundation; RQ3, RQ4
Karpukhin et al. (2020)	Dense passage retrieval	Dual-encoder dense retrieval (DPR)	Dense retrieval outperforms lexical retrieval for open-domain QA	Dense retriever design; RQ1
Robertson & Zaragoza (2009)	Probabilistic IR	BM25 ranking function	Establishes BM25 as a strong lexical baseline	BM25 baseline; RQ1, RQ2
Wang et al. (2022)	Text embeddings	Contrastive pre-training (E5)	Contrastive embeddings improve semantic retrieval	E5 retriever; RQ1, RQ2
Xiao et al. (2024)	General embeddings	Packaged embedding resources (BGE)	General-purpose embeddings perform strongly across domains	BGE retriever; RQ1, RQ2
Paul et al. (2023)	Legal LMs, Indian law	Domain-adaptive pre-training (InLegalBERT)	Legal pre-training improves Indian legal text representation	InLegalBERT retriever; RQ1, RQ2
Nigam et al. (2023)	Indian legal QA	Lexical + semantic retrieval with LLMs	Semantic retrieval outperformed lexical; expert evaluation needed	RQ1, RQ2; RQ3 rubric design
Pipitone & Houir Alam (2024)	Legal RAG evaluation	LegalBench-RAG benchmark	Retrieval quality strongly shapes legal answer quality	Benchmark design; RQ1–RQ3
Es et al. (2024)	RAG evaluation methodology	Automated evaluation (RAGAs)	Faithfulness/relevance can be evaluated systematically	Answer-quality rubric; RQ3
Magesh et al. (2024)	Commercial legal AI	Expert benchmarking	Commercial legal AI tools still hallucinate citations	Need for grounded legal AI; RQ3, RQ4

2.3 Thematic Synthesis

2.3.1 Legal NLP and retrieval methodology.

Legal natural language processing has progressed through successive methodological phases: rule-based and statistical approaches, neural and Transformer-based representation learning, and, most recently, retrieval and generation pipelines built on large language models (Hartung et al., 2023). Within the Indian legal context specifically, semantic retrieval has been found to yield measurable gains over lexical baselines, while expert evaluation of generated answers — rather than retrieval metrics alone — has been identified as necessary to establish real-world usefulness (Nigam et al., 2023). This line of work has since been extended through purpose-built Indian legal question-answering datasets (Koildurai Mishra, 2025) and long-context legal question-answering research (Kulkarni et al., 2026). Lexical ranking functions such as BM25 remain a strong and inexpensive baseline (Robertson Zaragoza, 2009), whereas dense dual-encoder methods outperform lexical approaches in settings where query–passage vocabulary mismatch is common (Karpukhin et al., 2020). Embedding-model design has since diversified into contrastively pre-trained general-purpose embeddings such as E5 (Wang et al., 2022) and BGE (Xiao et al., 2024), alongside domain-adaptive pre-training approaches such as InLegalBERT (Paul et al., 2023). Hybrid retrieval approaches, which fuse lexical and dense rankings via reciprocal rank fusion, combine exact-term precision with semantic recall; heterogeneous benchmarking work such as BEIR similarly demonstrates that no single retrieval paradigm dominates uniformly across domains (Thakur et al., 2021), reinforcing the rationale for evaluating lexical, dense, and hybrid retrieval side by side.

2.3.2 Retrieval-augmented generation, prompting, and evaluation.

RAG conditions a generative model's output on retrieved evidence to improve factual grounding (Lewis et al., 2020), an approach that aligns naturally with the traceability requirements inherent to legal reasoning. Benchmark-based work in the legal domain shows that retrieval quality substantially constrains achievable answer quality regardless of the generation model's capability (Pipitone Houir Alam, 2024), a finding that directly motivates treating retrieval strategy, rather than generation-model selection, as the primary experimental variable in the present study. Related work on prompt engineering demonstrates that the structure of a legal RAG prompt materially affects task performance (Sriram et al., 2026; Trautmann et al., 2022; Yu et al., 2022), informing the design of a citation-constrained, context-only generation prompt. Chunk boundaries have likewise been identified as a first-order determinant of downstream answer reliability rather than an incidental implementation detail (Bennani Moslonka, 2026; Bhat et al., 2025). Evaluating the resulting pipeline requires both retrieval-stage metrics grounded in established relevance-judgment methodology (Al-Maskari et al., 2008) and generation-stage metrics such as those proposed by RAGAs (Es et al., 2024).

2.3.3 Hallucination, faithfulness, and safe abstention.

Grounding generation in retrieved evidence reduces, but does not eliminate, hallucination. Corpora such as RAGTruth document that hallucinated content persists even when a retrieval context is supplied (Wu et al., 2024), and benchmarking of commercial legal AI tools confirms that this risk is not merely theoretical: such systems continue to hallucinate case citations despite marketing claims of grounded output (Magesh et al., 2024). Research on abstention mechanisms argues that a system's capacity to recognize evidence insufficiency and decline to answer when evidence is inadequate constitutes a core reliability property in its own right (Wen et al., 2024), a position echoed in broader reviews of trustworthy legal AI (Nguyen et al., 2025) and consistent with the legal-AI literature's treatment of a system's ability to justify or withhold a conclusion as central to its legitimacy (Richmond et al., 2023).

2.4 Comparison of Existing Approaches

Across these three thematic areas, the literature converges on several points of agreement while leaving certain questions unresolved. There is broad agreement that semantic and hybrid retrieval generally outperform purely lexical retrieval in adjacent, non-legal settings (Karpukhin et al., 2020; Nigam et al., 2023), and that retrieval quality constrains downstream generation quality rather than acting as an independent factor (Pipitone & Houir Alam, 2024). There is likewise agreement that hallucination persists as a risk even in retrieval-grounded systems, which motivates treating abstention as a necessary design property rather than an optional safeguard (Wu et al., 2024; Wen et al., 2024; Magesh et al., 2024).

Where the literature is less settled is in the direct comparison of embedding-model design philosophies. General-purpose contrastively trained embeddings (E5, BGE) and domain-adaptive pre-training (InLegalBERT) represent divergent strategies for improving dense retrieval, but, as detailed further in Section 2.4 below, no prior study had benchmarked these three specific models against one another on an Indian Supreme Court legal retrieval task at the time of this review. Their relative performance in this setting therefore remained an open empirical question rather than a resolved one. Similarly, while BEIR-style benchmarking establishes that no single retrieval paradigm dominates uniformly across domains in general (Thakur et al., 2021), the literature reviewed does not report a controlled, within-domain comparison of lexical, dense, and hybrid retrieval specifically for Indian Supreme Court legal question answering. Finally, although abstention and hallucination have each been studied independently, the reviewed literature does not report work that statistically connects retrieval-stage performance to both generation-stage answer quality and abstention reliability within a single evaluated system.

2.5 Embedding-Model-Specific Hypotheses

Because the present study compares three distinct dense embedding models—E5, BGE, and InLegalBERT—differing in training objective, training-data domain, and query/passage formatting convention, it is necessary to state explicitly, prior to evaluation, what performance differences were expected among them and why. Table 2

summarizes these pre-registered expectations, derived from each model's published training methodology rather than from prior empirical comparison in this specific domain.

Table 2 Pre-Registered Embedding Model Expectations

Model	Training Approach	Expected Behavior	Rationale
E5	Contrastive pre-training on large-scale, general-domain text-pair data, using an explicit query/passage prefix convention (`query:` / `passage:`) at inference (Wang et al., 2022).	Expected to perform well on general semantic similarity and retrieval, potentially matching or exceeding other general-purpose dense embedding models, but without explicit legal-domain adaptation.	The contrastive objective is designed to optimize embedding representations for retrieval-oriented similarity tasks. The prefix convention signals query or passage roles but does not provide explicit adaptation to legal terminology or citation structures.
BGE	General-purpose embedding model trained on broad, cross-domain corpora without an explicit query/passage prefix convention (Xiao et al., 2024).	Expected to perform comparably to E5, although differences may arise from variations in training data and objectives. Strong cross-domain retrieval performance was expected, but without legal-domain specialization.	BGE and E5 target general-purpose semantic retrieval through different training pipelines and data mixtures. In the absence of legal-domain training signals, substantial performance differences were not expected *a priori*.
InLegalBERT	Domain-adaptive pre-training of a BERT-family encoder on Indian legal text (Paul et al., 2023).	Expected to provide an advantage in representing Indian legal terminology and citation patterns, while potentially underperforming retrieval-specific dense models because it was not explicitly trained using a retrieval or contrastive objective.	Legal-domain pre-training may improve representations of statute names, citation formats, and domain-specific phrasing. However, without a retrieval-oriented objective comparable to those used by E5 or BGE, its embedding space may be less optimized for nearest-neighbor retrieval, potentially offsetting its domain-vocabulary advantage.

These expectations are stated at the level of directional reasoning rather than as predictions of specific metric values, since no prior study had benchmarked these three models on an Indian Supreme Court legal retrieval task; in particular, whether InLegalBERT's domain-vocabulary advantage would outweigh its lack of a retrieval-specific training objective was a genuinely open empirical question. Recording these hypotheses in advance of evaluation guards against hindsight bias in interpreting the retrieval results reported later in this study.

2.6 Relevance to the Present Study

The reviewed literature informed the present study's design at several levels. Methodologically, the demonstrated advantage of dense retrieval over lexical retrieval in general and adjacent legal settings (Karpukhin et al., 2020; Nigam et al., 2023), together with evidence that no single retrieval paradigm dominates uniformly across domains (Thakur et al., 2021), motivated a controlled, side-by-side evaluation of lexical, dense, and hybrid retrieval rather than adoption of a single presumed-superior method. Model selection was informed directly by the divergent training philosophies represented in the literature: BM25 as an established lexical baseline (Robertson Zaragoza, 2009); E5 and BGE as contrastively trained general-purpose dense embeddings (Wang et al., 2022; Xiao et al., 2024); and InLegalBERT as a domain-adaptive alternative (Paul et al., 2023), combined with a hybrid BM25–E5 configuration reflecting the reciprocal-rank-fusion approach to combining lexical and semantic signal.

Benchmark construction was shaped by the finding that retrieval quality substantially constrains legal answer quality (Pipitone Houir Alam, 2024) and by the observation that expert evaluation, rather than retrieval metrics alone, is necessary to establish real-world usefulness in the Indian legal QA setting (Nigam et al., 2023); this motivated building the benchmark from questions and relevance judgments contributed directly by practicing judicial officers. The generation pipeline's citation-constrained, context-only prompt design draws on evidence that legal RAG prompt structure materially affects task performance (Sriram et al., 2026; Trautmann et al., 2022; Yu et al., 2022) and on RAG's general premise of conditioning generation on retrieved evidence to improve factual grounding (Lewis et al., 2020). The evaluation strategy combines retrieval-stage metrics grounded in established relevance-judgment methodology (Al-Maskari et al., 2008) with generation-stage metrics in the spirit of RAGAs (Es et al., 2024). Finally, the study's treatment of abstention as a primary, separately evaluated reliability property, rather than a secondary consideration, is informed directly by evidence that hallucination persists even in retrieval-grounded systems (Wu et al., 2024; Magesh et al., 2024) and by the argument that the ability to recognize evidence insufficiency is itself a core reliability property (Wen et al., 2024; Nguyen et al., 2025; Richmond et al., 2023).

2.7 Research Gap

The literature reviewed above establishes strong individual foundations in the evolution of legal NLP, retrieval methodology, RAG architecture, and hallucination and abstention research. However, no existing study identified in this review combines

these threads into a single controlled evaluation for Indian Supreme Court legal question answering. Specifically, prior work does not (a) compare lexical, dense, and hybrid retrieval within one experimental design using a shared corpus, chunking scheme, and benchmark; (b) use a benchmark built from questions and relevance judgments contributed directly by practicing judicial officers rather than crowd-sourced or synthetic queries; or (c) statistically connect retrieval-stage performance to both generation-stage answer quality and abstention reliability within the same evaluated system. The present study addresses this gap by evaluating six retrieval configurations under controlled conditions and validating the resulting differences using Wilcoxon signed-rank tests, Friedman tests with Holm-corrected post-hoc comparisons, and exact binomial tests.

3 Materials and Methods

3.1 Study Overview and Workflow

This study examines how the choice of retrieval strategy affects a retrieval-augmented generation (RAG) system (Lewis et al., 2020) that answers legal questions from Indian Supreme Court judgments (the system is referred to as NyayaRAG below). Six retrieval configurations were compared within one pipeline: TF-IDF, BM25, three dense embedding models (E5, BGE and InLegalBERT), and a hybrid retriever that fuses the BM25 and E5 rankings by Reciprocal Rank Fusion (RRF). Together these span lexical, dense and hybrid approaches (Manning et al., 2008). Retrieval strategy was designated the primary experimental factor because retrieval quality constrains the answer quality that a legal RAG system can reach, whatever generation model is used (Pipitone Houir Alam, 2024). The strongest retriever was subsequently combined with a large language model to produce citation-constrained answers, and the resulting system was evaluated for answer quality and for its ability to abstain when the retrieved evidence was insufficient.

The corpus, chunking scheme, benchmark questions, retrieval depth (k = 5) and generation temperature (0.0) were identical across all conditions, so that retrieval strategy was the only manipulated variable.The work proceeded in seven stages (Fig. 1):

1. Corpus acquisition, text extraction, cleaning, and paragraph-aware chunking. Data preparation. 782 judgments from 2024 were extracted, cleaned and normalized, then segmented by a paragraph-aware chunker (configuration $\text{chunk}_3 84_4 0) into 30,521 chunks$.

1. Index construction. Six retrieval configurations were built over the shared chunk corpus.
2. Benchmark construction and relevance annotation. Questions from 29 judicial officers gave 135 candidate questions. The top five chunks returned by each model for each question were annotated manually, and question-level answerability was derived from these annotations, yielding 60 Answerable and 60 Unanswerable evaluation questions.

3. Retrieval evaluation (RQ1, RQ2). Precision@5, Recall@5, Hit Rate@5, MRR and nDCG@5 were computed. RQ1 was tested with a one-sided Wilcoxon signed-rank test and RQ2 with a Friedman test followed by Holm-corrected pairwise comparisons; the best retriever was then selected.
4. Evidence-grounded generation. The top five chunks and a citation-constrained prompt were given to Google Gemini, which either returned a cited answer or abstained.
5. Answer and abstention evaluation (RQ3, RQ4). Answers were scored on a five-dimension rubric and tested with a one-sample Wilcoxon test (RQ3); abstention was summarized in a confusion matrix and tested with an exact one-sided binomial test (RQ4).
6. Deployment. The validated pipeline was exposed through a Gradio conversational interface.

RQ1 and RQ2 were answered at stage 4. RQ3 and RQ4 were answered at stage 6, using the configuration selected at stage 4 and carried unchanged into stage 5. The deployed system at stage 7 reuses that configuration without a separate evaluation.

Figure 1
NyayaRAG End-to-End Workflow: Data Preparation, Retrieval, Generation, and Evaluation

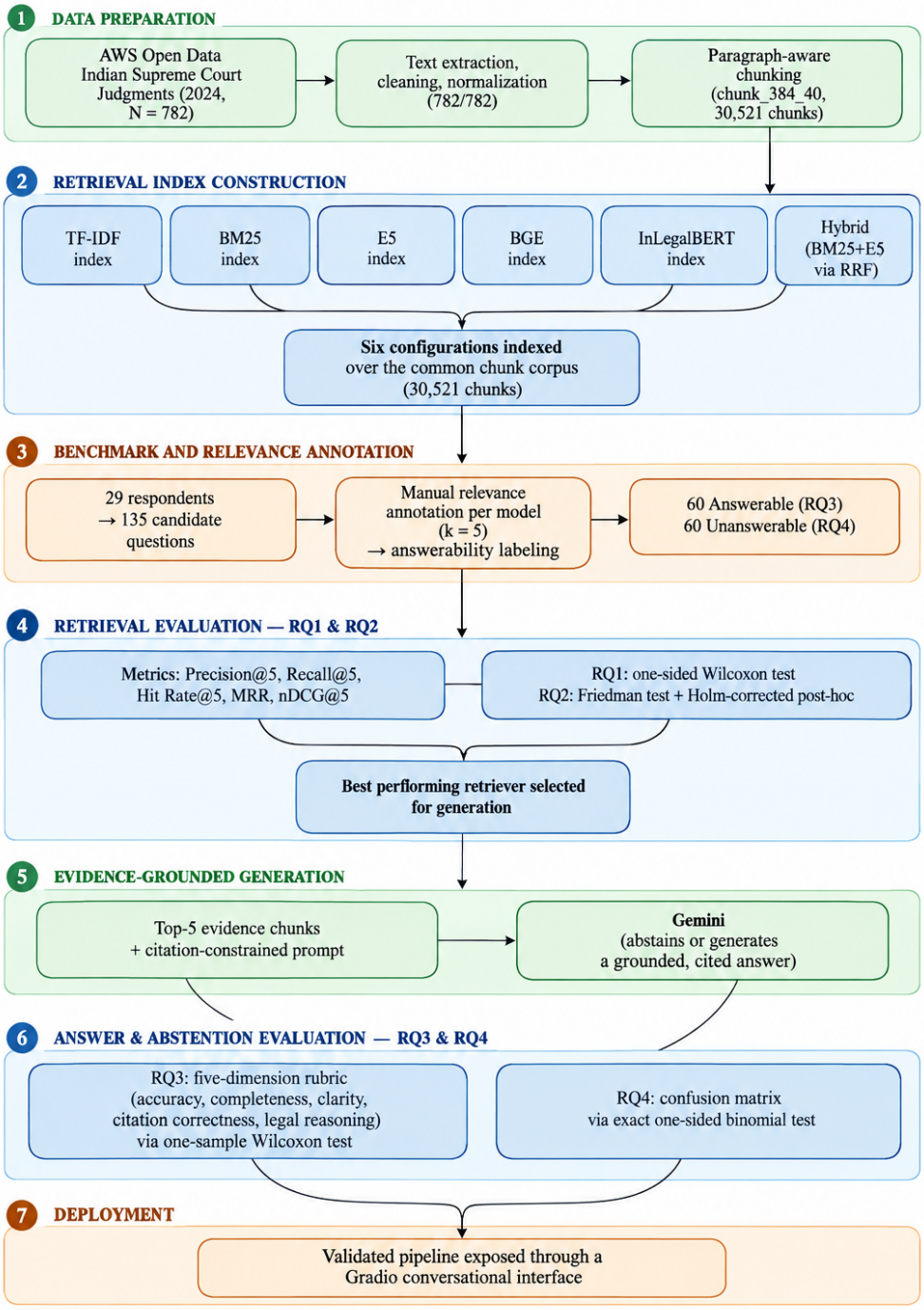


Fig. 1 Seven-stage workflow. 1 — Data Preparation: AWS Open Data Indian Supreme Court Judgments (2024, N = 782) → extraction, cleaning, normalization (782/782) → paragraph-aware chunking $(\text{chunk}_384_40, 30,521 chunks).2|Retrieval Index Construction : six configurations indexed over the common chunk corpus|TF-IDF, BM25, E5, BGE, InLegalBERT, Hybrid(BM25+E5 via RRF).3|Benchmark and Relevance Annotation : 29 respondents ß 135 candidates ß manual relevance annotation per model(k = 5) ß answerability labeling ß 60 Answerable(RQ3)/60 Unanswerable(RQ4).4|Retrieval Evaluation(RQ1/RQ2) : Precision@5, Recall@5, Hit Rate@5, MRR, nDCG@5; RQ1 one - sided Wilcoxon, RQ2 Friedman + Holm - corrected post - hoc; best retriever selected.5|Evidence - Grounded Generation : top - 5 evidence+citation-constrained prompt ß Gemini, which abstains or generates a grounded, cited answer.6|Answer/Abstention Evaluatio RQ3 five - dimension rubric via a one - sample Wilcoxon; RQ4 confusion matrix via exact one - sided binomial test.7|Deployment : the validated pipeline exposed through a Gradio conversational interface. Source : workflow_diagram.png.$

3.2 Data Source and Acquisition

Judgment text and metadata came from the Indian Supreme Court Judgments dataset in the AWS Open Data registry (Indian Supreme Court Judgments, 2024). The registry mirrors judgment documents and case metadata originally published through the eCourts Services platform. Only the English-language partition for the year 2024 was used. This restriction to a single year and a single language was a deliberate scoping decision, intended to validate the full pipeline before extending it to a larger corpus.

The partition contained 782 judgment PDFs and a metadata table of 782 records. After filename normalization every PDF matched exactly one record, with no missing or surplus files on either side. All 782 judgments were retained, and no judgment was excluded at any later stage of preparation.

3.3 Dataset Description

3.3.1 Judgment corpus and metadata.

Each metadata record held 18 fields from the source registry, which fall into four groups: case identification, adjudication, descriptive summary, and provenance or technical information (Table 3). The field `decision_date` was converted to a datetime type (format `%d-%m-%Y`, with invalid entries coerced), and `year` was converted from string to integer. All other fields were held as strings.

Table 3 Raw metadata fields ($N = 782$ records) and percentage of missing values (empty strings plus nulls).

Field	Group	Content	% missing
title	Case identification	Case or judgment title as recorded by the registry	0.00
petitioner	Case identification	Petitioner or appellant name(s)	0.77
respondent	Case identification	Respondent name(s)	0.77
citation	Case identification	Neutral or official citation string	0.00
case_id	Case identification	Registry-internal case identifier	4.73
cnr	Case identification	Case Number Record (CNR) identifier	0.00
judge	Adjudication	Presiding judge(s) or bench	0.51
author_judge	Adjudication	Judge who authored the opinion	100.00
decision_date	Adjudication	Date of decision	0.00
disposal_nature	Adjudication	Nature of disposal (e.g., allowed, dismissed)	10.74
court	Adjudication	Court identifier (Supreme Court of India)	0.00
description	Descriptive	Free-text case summary supplied by the registry	79.28
nc_display	Descriptive	Display-formatted neutral citation	4.73
raw_html	Descriptive	Raw HTML snippet of the registry listing	0.00
path	Provenance	Filesystem stem linking a record to its PDF	0.00
available_languages	Provenance	Languages in which the judgment is available	20.72
scraped_at	Provenance	Timestamp of collection from the registry	0.00
year	Provenance	Judgment year (2024 for all records)	0.00

3.3.2 Derived datasets.

Preparation produced a processed judgment table of 782 rows and 19 columns. Seven of its columns were derived during preparation: `raw_text` (text extracted from the PDF), `clean_text` (output of the cleaning function), `judgment_text` (cleaned text truncated at the start of the judgment proper, used for all chunking, embedding, and retrieval), `sentence_count`, `paragraph_count`, `avg_words_per_paragraph`, and `lexical_diversity`. The chunk corpus used for every retrieval model had 30,521 rows and nine columns: `global_chunk_id`, `judgment_id`, `case_title`, `configuration`, `local_chunk_id`, `chunk_text`, `word_count`, `paragraph_count`, and `split_paragraph_count`.

3.3.3 Evaluation datasets

Evaluation used questions contributed by judicial officers (Section 9.2). Table 2 lists the datasets and their sizes.

Table 4 Corpus and evaluation datasets.

Dataset	Size	Use
Judgment corpus (2024, English)	782 judgments	Source of retrievable text
Chunk corpus (`chunk_384_40`)	30,521 chunks (39.03 per judgment)	Shared index for all six retrievers
Raw benchmark questions	145 (29 respondents × 5)	Question pool
Pilot questions	10	Pipeline validation; excluded from final evaluation
Candidate benchmark	135 questions	Retrieval evaluation (RQ1, RQ2); $N = 134$ for Recall@5
RQ3 evaluation set	60 Answerable questions	Answer-quality evaluation
RQ4 evaluation set	60 Unanswerable questions	Abstention evaluation

3.4 Data preprocessing and cleaning

3.4.1 Text extraction

Text was extracted from each PDF with PyMuPDF 1.28.0 using `page.get_text("text", sort=True)`. Extraction succeeded for all 782 files. Validation of the extracted text files confirmed that all 782 expected files were present and that none was empty, none fell below a 100-word completeness threshold, and none produced a read error. Raw word counts across the 782 judgments had a mean of 8,725.34 (SD 13,521.21), a minimum of 320, a first quartile of 3,179.75, a median of 5,244.50, a third quartile of 8,805.25, and a maximum of 144,445.

3.4.2 Missing values

Missingness was quantified per metadata field (Table 3). `author_judge` was missing in every record (100.00%) and was excluded from downstream analysis without imputation. `description` was missing in 79.28% of records and was kept as a supplementary field of secondary importance. The fields needed to build the corpus and to ground citations (`title`, `path`, `citation`, `court`, `decision_date`, and `year`) had no missing values.

3.4.3 Duplicate handling

The duplicate audit found no exact duplicate rows, no repeated citation values, and no repeated path values. A default duplicate check on `case_id` flagged 39 rows in two groups. In the first group, 37 rows had an empty `case_id`; a standard pandas duplicate check treats empty strings as equal, so these rows were an artifact of blank identifiers and not identifier collisions. Each of them remained identifiable through its path, title, and citation. The second group was a single genuine collision: the value "2024 INSC 293" had been assigned by the registry to two different judgments decided one

day apart (paths `2024_4_576_590`, decided 2024-04-09, and `2024_4_357_371`, decided 2024-04-10). This was traced to the source metadata, and both judgments remained distinguishable by path, title, and citation. No rows were removed as a result of the duplicate audit.

3.4.4 Text normalization

A custom function, `clean_legal_text()`, corrected layout artifacts introduced by PDF extraction. It normalized line endings, removed tabs and leading or trailing whitespace on each line, rejoined words that had been split by a hyphen at a line break, replaced single line breaks with spaces while restoring genuine paragraph breaks, and collapsed repeated spaces. The operation was confined to formatting. No stop words, punctuation, numbers, or citations were removed. Across the 782 judgments, cleaning removed on average 8,933.60 characters but only 3.44 words per judgment (Table 5).

Table 5 Text statistics before and after cleaning ($N = 782$ judgments).

Measure	Mean	SD	Min	Q1	Median	Q3	Max
Raw characters	61,353.48	98,372.57	2,575	21,708.50	36,121.50	61,454.50	1,029,556
Clean characters	52,419.88	82,404.42	2,032	18,951.25	31,398.00	52,681.00	903,703
Raw words	8,725.34	13,521.21	320	3,179.75	5,244.50	8,805.25	144,445
Clean words	8,721.90	13,516.21	320	3,174.25	5,240.50	8,802.50	144,369
Characters removed	8,933.60	18,066.21	543	2,779.00	4,832.00	8,599.50	320,483
Words removed	3.44	6.02	0	1.00	2.00	4.00	76

The cleaned corpus was saved as `clean_corpus.parquet` (782 rows × 15 columns).

3.4.5 Removal of editorial front matter

After cleaning and before chunking, editorial material at the head of each judgment (citation blocks, issue summaries, headnotes, counsel appearances, and page metadata) was removed. Each cleaned text was truncated to begin at the first occurrence of the marker string `Judgment / Order of the Supreme Court`, with the marker retained; texts lacking the marker were left unchanged. On the sample judgment used to check this step, text length fell from 42,201 to 36,436 characters, a reduction of 5,765 characters. Corpus-wide validation found 782 judgments with no missing and no empty `judgment_text`.

3.4.6 Data stages

Table 6 Dataset dimensions at successive stages of the pipeline.

Stage / artifact	Dimensions	Note
Raw metadata (`metadata_2024.parquet`)	782 rows × 18 columns	Registry export
Cleaned corpus (`clean_corpus.parquet`)	782 rows × 15 columns	After `clean_legal_text()`
Processed corpus (`metadata_2024_processed.parquet`)	782 rows × 19 columns	After derived analytical columns were added
Chunk corpus (`chunks_384_40.parquet`)	30,521 rows × 9 columns	Final chunk configuration
PDF-to-metadata correspondence	782 PDFs ↔ 782 records	No missing or surplus PDFs
Text extraction	782 / 782 files	No failed extractions, empty files, files under 100 words, or read errors
Judgments excluded	0	No rows dropped at any stage

3.4.7 Exploratory data analysis

The exploratory analysis reported here is limited to corpus properties that shaped the design of the chunking scheme and the retrieval evaluation.

Judgment length. After cleaning, judgments averaged 8,721.90 words with a median of 5,240.50 (range 320 to 144,369). The distribution was strongly right-skewed (Fig. 2). By the interquartile-range rule (Q1 = 3,174.25, Q3 = 8,802.50, IQR = 5,628.25, upper bound 17,244.88 words), 73 judgments (9.34 %) were length outliers; the lower bound was negative, so no low outliers were possible. Many judgments therefore far exceed the 512-token input limit of the dense encoders. Indexing whole judgments was consequently infeasible, and chunk-level indexing was adopted throughout.

Paragraph and sentence structure. Paragraphs were delimited by blank lines. The corpus contained 47,258 paragraphs and 342,405 sentences (Table 5). Despite wide variation in judgment length, the per-judgment mean paragraph length was stable, at 140.55 words (SD 22.56; median 141.90; range 41.98 to 203.10), with most values between 120 and 160 words. The corpus-wide ratio of total words to total paragraphs

was 144.33; this differs from the mean of per-judgment ratios because the two statistics are computed differently. The consistency of paragraph length indicates that the reasoning in these judgments is organized in paragraphs of similar size. This informed the candidate chunk sizes of 256, 384 and 512 words and supported chunk targets of 384 to 512 words, which correspond to roughly three to five paragraphs.

Table 7 Structural statistics per judgment ($N = 782$).

Variable	Total	Mean	SD	Median	Min	Max	Q1	Q3
Sentences	342,405	437.86	671.18	273.00	35	7,846	164.25	448.75
Paragraphs	47,258	60.43	105.60	37.00	6	1,773	24.00	59.00
Average words per paragraph	n/a	140.55	22.56	141.90	41.98	203.10	126.96	156.37
Lexical diversity (unique / total tokens)	n/a	0.1799	0.0551	0.1774	0.0460	0.4856	0.1444	0.2113

Vocabulary and content. Lexical diversity computed on the pooled corpus was 0.0074 (47,961 unique tokens among 6,492,583 tokens), a low value that is expected in a large and repetitive legal corpus. The per-judgment average in Table 7 is not comparable with this pooled ratio because the two are computed at different levels of aggregation. A curated list of 25 legal terms was counted case-insensitively across the corpus. The most frequent were *court* (59,271), *section* (30,034), *act* (28,406), *state* (23,109), *case* (20,792), and *order* (19,350), and the list included criminal-litigation terms (for example, *appellant*, *accused*, *offence*) as well as constitutional and public-law terms (for example, *article*, *rights*). Statute mentions were counted after mapping abbreviations and alternative names onto 44 canonical statute names. The Constitution of India was cited most often (7,884), followed by the Code of Criminal Procedure (2,928) and the Indian Penal Code (2,641) (Fig. **??**). Taken together, these counts indicate coverage of criminal, constitutional, and civil matters suited to a legal question-answering benchmark.

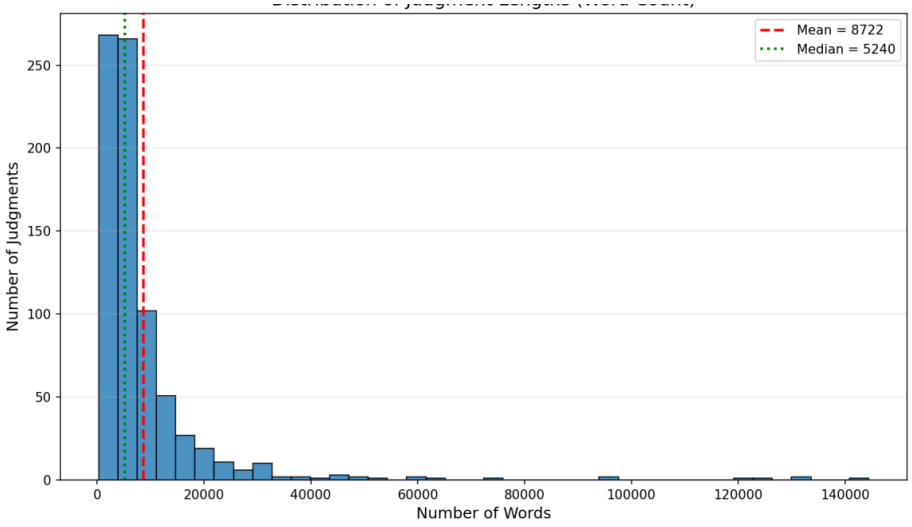


Fig. 2 Histogram of judgment word counts (40 bins), with reference lines at the mean (8,722) and median (5,240), showing a strongly right-skewed distribution. Source: NB01 media file fig2.png

3.4.8 System architecture

The system converts a natural-language legal question into either an answer that cites its supporting chunks or a fixed abstention statement. Table 6 gives the input, processing, output and technology of each component. The chunker, retrievers and generator are specified further in Sections 7 and 8.

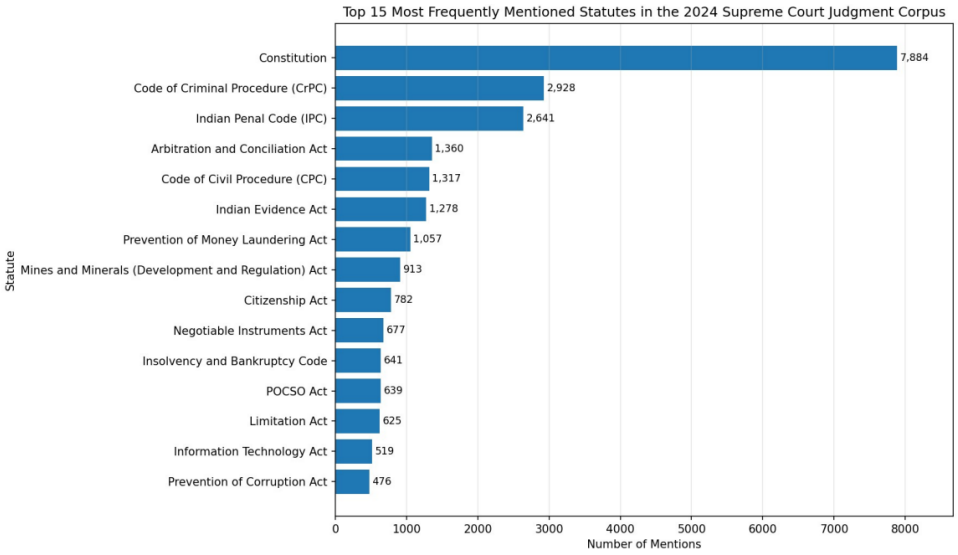


Fig. 3 Horizontal bar chart of the top 15 statutes by canonical citation frequency, led by the Constitution of India (7,884), Code of Criminal Procedure (2,928), and Indian Penal Code (2,641). Source: NB01 media file fig3.png.

Table 8 Pipeline components.

Component	Input	Processing	Output	Technology
Extraction	782 judgment PDFs	Text extraction and file-level validation	Raw judgment text	PyMuPDF 1.28.0
Cleaning and front-matter removal	Raw text	Formatting normalization; truncation at the judgment marker	`judgment_text`	Custom `clean_legal_text()` and marker rule
Chunking	`judgment_text`	Paragraph-aware segmentation (target 384 words, overlap 40 words)	30,521 chunks	Custom chunker
Indexing	Chunk corpus	TF-IDF matrix; BM25 index; E5, BGE, and InLegalBERT embeddings	Six retrievable indices (Hybrid reuses the BM25 and E5 indices)	scikit-learn, `rank_bm25`, `sentence-transformers`, ChromaDB 1.5.9
Retrieval	Question	Top-5 ranking per configuration; Hybrid fuses the top 20 of BM25 and E5 by RRF	Five ranked chunks	As above
Prompt assembly and generation	Question and five evidence blocks (case title, judgment ID, chunk ID, chunk text)	Citation-constrained, evidence-only prompt at temperature 0.0	Cited answer, or the fixed abstention sentence	Google Gemini (`gemini-3.6-flash`)
Behaviour classification (evaluation only)	Generated text	Detection of the abstention sentence; manual safety annotation for Unanswerable questions	Answered / Abstained labels	Substring matching; manual rubric
Deployment	User turn and up to five prior turns	Fresh Hybrid retrieval per turn, grounded generation, citation display	Answer or abstention, with a source list	Gradio 6.20.0

Two rules define the behaviour of the generation stage. First, the legal content of an answer may come only from the retrieved evidence supplied for that question. Second, abstention is decided by the generation model under the prompt instruction alone; no retrieval score or confidence threshold is involved.

Deployment interface. The deployed application implements the sequence user query, Hybrid retrieval, top-5 evidence, grounded generation, citation and evidence display, and answer or abstention. It uses the Hybrid BM25–E5 configuration and the frozen generation prompt without re-evaluation. In the multi-turn interface, each turn triggers a new Hybrid retrieval that is not cached across turns, and a lightweight follow-up-aware constructor prepends the preceding user turn to the query whenever conversation history exists. Up to five prior turns are passed to the generator to resolve references in follow-up questions, but the prompt restricts the legal content of each answer to the evidence retrieved for the current turn. The generator settings match those of the frozen evaluation (`gemini-3.6-flash`, temperature 0.0, top-p 1.0), except that the maximum output allowance was raised from 1,200 to 2,000 tokens to limit truncation of longer conversational answers. The abstention sentence is unchanged, and the model is instructed not to expose retrieval diagnostics to the user. The interface was launched with `demo.queue().launch(share=True, debug=False)`, which creates a temporary public link valid for up to one week. No latency or hardware benchmarking was carried out.

Citation backfill. The chunk corpus initially lacked a formal citation column. It was established that `judgment_id` maps exactly to the judgment-level citation field across all 782 unique judgments, and this key was used to join formal citations (for example, "[2024] 2 S.C.R. 943") onto retrieved chunks. Deduplicated authorities are listed under a "Sources" heading, formatted as case title followed by formal citation. Because the backfill was built for the deployed interface, the evidence supplied during the RQ3 and RQ4 evaluations carried only the case title and judgment identifier.

3.5 Chunking

3.5.1 Approach and candidate configurations

Chunk boundaries affect the reliability of downstream question answering (Bennani & Moslonka, 2026; Bhat et al., 2025). Since indexing whole judgments was infeasible (Section 5), each judgment was divided into overlapping chunks by a custom Universal Paragraph-Aware Chunker that respects paragraph boundaries. Chunk size and overlap were measured in words, counted with the regular expression `\b\w+\b`, and not with a model-specific subword tokenizer. Six candidate configurations were defined by crossing three target sizes with two overlap ratios (Table 9).

Table 9 Candidate chunk configurations.

Configuration	Target size (words)	Overlap (words)	Overlap (%)
chunk_256_26	256	26	10
chunk_256_51	256	51	20
chunk_384_40	384	40	10
chunk_384_80	384	80	20
chunk_512_50	512	50	10
chunk_512_100	512	100	20

3.5.2 Algorithm

For each judgment the chunker applied four steps.

1. **Front-matter removal.** The cleaned text was truncated at the first occurrence of the marker "Judgment / Order of the Supreme Court" (Section 4.5).
2. **Paragraph segmentation.** The text was split at blank lines (`re.split(r"\n\n", text)`), and empty segments were discarded.
3. **Chunk assembly with a tolerance bound.** Paragraphs were appended in order until the running word count reached the target size. The maximum permitted size was `int(1.20 × target size)`. If adding the next paragraph would push a non-empty chunk beyond this maximum, the chunk was closed before that paragraph, so a chunk was finalized early and did not overshoot. The first paragraph of a chunk was always accepted, which guarantees forward progress. A single paragraph longer than the target size was divided into segments of target-size words, and each segment was handled as its own paragraph.
4. **Overlap carry-forward.** The next chunk started at the first paragraph of the shortest run of whole trailing paragraphs of the closed chunk whose combined length reached the overlap size, provided that this start fell strictly after the start of the previous chunk. Overlap was therefore built from complete paragraphs and never from a mid-paragraph word split. Only paragraphs that individually exceeded the target size were split at word level.

Algorithm 1 Universal Paragraph-Aware Chunking

Require: Cleaned judgment text, target chunk size, overlap size
Ensure: Ordered list of chunks for the judgment

Cut the text to start at the judgment marker, or keep it unchanged if the marker is absent
Split the text into paragraphs at blank lines and discard empty paragraphs
Set the maximum chunk size to 1.2× target size
while paragraphs remain to be chunked **do**
    Start a chunk at the current start paragraph
    for each following paragraph **do**
        if the paragraph is longer than the target size **then**
            Split it into target-size word segments and treat each as a paragraph
        end if
        if adding the paragraph would exceed the maximum size and the chunk is not empty **then**
            End the chunk before this paragraph
        else
            Add the paragraph to the chunk
            if the chunk has reached the target size **then**
                End the chunk after this paragraph
            end if
        end if
    end for
    Save the chunk
    Find the fewest whole trailing paragraphs whose words reach the overlap size
    Set the next start to the first of those paragraphs, but strictly after the previous start
end while

3.5.3 Development and validation of the chunker

The chunker was refined in stages on a single sample judgment before being applied to the corpus. The initial version had no tolerance bound and added paragraphs until the target was met, which allowed unbounded overshoot. Front-matter removal alone reduced the number of chunks at every target size (for `chunk_384_40`, from 30 to 28), and adding the 20% tolerance bound then lowered the maximum chunk length at every target size (for `chunk_384_40`, from 687 to 454 words). It also raised the chunk count, because oversized chunks were now closed earlier (for `chunk_384_40`, from 28 to 29). The refined version, which includes both changes, was applied to all 782 judgments.

Corpus-wide structural statistics for the six configurations are given in Table 10.

Table 10 Corpus-wide chunking statistics by configuration.

Configuration	Total chunks	Mean words	Median words	Min words	Max words	Mean paragraphs/chunk	Split paragraphs
chunk_256_26	45,971	219.56	256.0	7	307	1.99	45,280
chunk_256_51	46,075	219.51	256.0	7	307	2.00	45,280
chunk_384_40	30,521	363.35	381.0	15	460	2.77	14,507
chunk_384_80	31,169	363.35	381.0	15	460	2.78	14,507
chunk_512_50	27,233	401.89	393.0	60	614	3.09	94
chunk_512_100	27,868	402.11	393.0	60	614	3.09	94

3.5.4 Final configuration and rationale

`chunk_384_40` (target 384 words, overlap 40 words) was selected. The 256-word configurations produced the most chunks and the highest incidence of split paragraphs, whereas the 512-word configurations preserved nearly every paragraph boundary but yielded larger units. The 384-word configuration was judged to offer the best balance of granularity, coherence, and efficiency. The choice rested on these structural properties and on the paragraph-length analysis in Section 5; retrieval performance was not compared across chunk configurations. The configuration was applied identically to all retrievers.

The final corpus contained 30,521 chunks from 782 judgments (39.03 per judgment), with chunk lengths of 15 to 460 words (mean 363.35, median 381.00). Validation found no empty chunks, no duplicate chunk texts (`chunk_text.duplicated().sum()` = 0), 30,521 unique `global_chunk_id` values, and no missing chunk text, judgment identifiers, or case titles. Two manual checks were also run. In the first, the first three chunks of the judgment [2024] 10 S.C.R. 108 (396, 384, and 421 words, each spanning three paragraphs) were read to confirm continuity across chunk boundaries. In the second, the first chunk of a separate sample judgment was inspected and found to retain the case citation, title, judicial authorship, paragraph numbering, party names, and statutory references.

3.6 Model and algorithm configuration

3.6.1 Retrieval models

All retrievers operated on the shared `chunk_384_40` corpus (30,521 chunks). Table 11 lists the configuration of each model. None of the embedding models was fine-tuned on the corpus or the benchmark.

Table 11 Retrieval model configurations.

Model	Implementation	Settings	Representation and index	Similarity
TF-IDF (Salton & Buckley, 1988)	scikit-learn `TfidfVectorizer`	`lowercase=True`, `stop_words=None`, `ngram_range=(1,2)`, `min_df=5` (all set explicitly); `max_df=1.0`, `norm="l2"`, `use_idf=True`, `smooth_idf=True` (library defaults)	Sparse matrix of 30,521 chunks × 230,196 features with 12,406,627 non-zero entries	Cosine similarity
BM25 (Robertson & Zaragoza, 2009)	`rank_bm25.BM25Okapi`	Lowercase whitespace tokenization; no stemming, lemmatization, or stop-word removal; `k1` and `b` not set (library defaults, 1.5 and 0.75 per the package documentation)	30,521 tokenized chunks; mean chunk length 363.35 words; 11,089,911 indexed tokens	BM25 Okapi scori
E5 (Wang et al., 2022)	`intfloat/e5-base-v2` via sentence-transformers	Prefix "query: " for queries and "passage: " for chunks; `normalize_embeddings=True`; batch size 32; maximum sequence length 512 tokens (model default)	768-dimensional L2-normalized embeddings in ChromaDB collection `e5_collection`	Cosine (`hnsw:spa` = cosine)
BGE (Xiao et al., 2024)	`BAAI/bge-base-en-v1.5` via sentence-transformers	No query or passage prefix (raw text encoded); `normalize_embeddings=True`; batch size 32; maximum sequence length 512 tokens	768-dimensional L2-normalized embeddings in `bge_collection`	Cosine
InLegalBERT (Paul et al., 2023)	`law-ai/InLegalBERT` via sentence-transformers	No prefix; pooling inherited from the configuration stored on the Hugging Face Hub (presumed mean pooling, not verified); `normalize_embeddings=True`; batch size 32; maximum sequence length 512 tokens	768-dimensional L2-normalized embeddings in `inlegalbert_collection`	Cosine

Further implementation details follow.

- **TF-IDF.** The vectorizer was fitted on all 30,521 chunks. After the artifacts were saved and reloaded, the matrix shape, vocabulary size, and count of non-zero entries were identical to those before saving. Cosine similarity was the intended ranking measure, applied in the evaluation stage.
- **BM25.** The index was built by passing only the tokenized corpus to the constructor. Reloading the persisted index reproduced identical top-ranked results and scores.
- **Dense models.** Embeddings were inserted into ChromaDB persistent collections in batches of 500, with cosine distance specified at collection creation. Every collection was validated by asserting that the indexed count equalled the number of chunks. For E5 and BGE, the minimum, maximum, and mean L2 norm over all 30,521 embeddings was 1.000000, with no NaN or infinite values. For E5, queries were encoded with the "query: " prefix, mean pooling of the last hidden state, and L2 normalization. Each model's settings were saved to a JSON file (`e5_config.json`, `bge_config.json`, `inlegalbert_config.json`). The InLegalBERT load log listed unused masked-language-model and next-sentence-prediction head weights as unexpected, which is expected when only the encoder is used for sentence embeddings.

3.6.2 Hybrid BM25–E5 retriever

The Hybrid retriever draws on the BM25 and E5 indices described above and creates no new index. For each question, BM25 and E5 each returned their own top 20 chunks (`COMPONENT_TOP_K` = 20). The two candidate lists were joined by an outer merge on `global_chunk_id`, so that a chunk present in both lists appeared once. Each chunk received a Reciprocal Rank Fusion score:

$$\mathrm{RRF}(d) = \frac{1}{k_{\mathrm{RRF}} + r_{\mathrm{BM25}}(d)} + \frac{1}{k_{\mathrm{RRF}} + r_{\mathrm{E5}}(d)}, \tag{1}$$

where $r_{\mathrm{BM25}}(d)$ and $r_{\mathrm{E5}}(d)$ are the 1-indexed ranks of chunk d in the respective component lists, $k_{\mathrm{RRF}} = 60$ (described in the implementation as a conventional value), and a chunk absent from one list contributes 0 for that term. The two components are weighted equally. With $k_{\mathrm{RRF}} = 60$, a rank-1 result adds $1/61 \approx 0.016393$ to a component term and a rank-5 result adds approximately 0.015385. Chunks were sorted by fused score in descending order. Ties were broken first by "best component rank", the smaller of the chunk's BM25 and E5 ranks, in ascending order, and then by `global_chunk_id` in ascending order. The five highest-ranked chunks were retained as the final output (`HYBRID_TOP_K` = 5) and assigned ranks 1 to 5. The implementation was checked by confirming that each of the 135 evaluation questions returned exactly five results with ranks 1 to 5 (675 rows in total). No tuning or ablation over k_{RRF} or the component weights was performed, and the fused retriever was evaluated as one fixed configuration.

The components were chosen from the earlier evaluation of the individual models. BM25 was selected as the lexical component because it scored higher than TF-IDF in the descriptive comparison, and E5 was selected as the dense component because it had the best mean rank (1.0) among the three dense models on Precision@5, Hit

Rate@5, MRR, and nDCG@5. Neither the BM25–TF-IDF difference nor the E5–BGE difference was statistically significant in pairwise testing (all $p > 0.05$), so the selection rested on descriptive ranking.

3.6.3 Generation model and prompt

Generation used the Google Gemini API through the `google.genai` SDK. Table 12 gives the settings of the frozen RQ3 and RQ4 evaluation, which were identical for both question sets. One model, one prompt, and one decoding configuration were evaluated; no other generator was compared.

Table 12 Generation configuration for the RQ3 and RQ4 evaluations.

Parameter	Value
Provider	Google Gemini API
Model	`gemini-3.6-flash`
Temperature	0.0
Top-p	1.0
Maximum output tokens (frozen evaluation)	1,200
Retrieval model	Hybrid BM25–E5
Evidence per question (top-k)	5
Prompt version	`RQ3_Grounded_v1` / `RQ4_Grounded_v1` (identical content)

The prompt structure follows prior work showing that the framing of a legal prompt affects task performance (Sriram et al., 2026; Trautmann et al., 2022; Yu et al., 2022). It supplies the question together with five evidence blocks, each holding the case title, judgment identifier, chunk identifier, and chunk text. The model is told to answer strictly from the supplied evidence, not to use legal knowledge outside it, not to make claims the evidence does not support, and not to invent case names, citations, or statutory provisions. It is asked to give a direct answer, the supporting legal principle or holding, and the case citation wherever the evidence provides one. Citations are copied verbatim from the evidence; no automated citation verification was applied, and citation accuracy was judged manually (Section 9.6). When the evidence is insufficient, the model must return the fixed sentence "The retrieved evidence is insufficient to provide a reliable answer." This is the only abstention mechanism in the system. Classification of a response as an abstention was later carried out by detecting this sentence in the generated text (Section 9.7).

3.7 Experimental design

3.7.1 Variables and design

The evaluation used a repeated-measures design in which the question was the unit of analysis: every benchmark question was processed by every retrieval model and, for RQ3 and RQ4, by a single generation pipeline. In the retrieval comparison (RQ1,

RQ2), the manipulated variable was the retrieval strategy, with six levels (TF-IDF, BM25, E5, BGE, InLegalBERT, and Hybrid BM25–E5). RQ1 additionally grouped the individual models into a lexical family (TF-IDF, BM25) and a semantic family (E5, BGE, InLegalBERT), defined before the results were examined. The Hybrid retriever belonged to neither family.

The following were held constant: the corpus (782 judgments), the chunk configuration (`chunk_384_40`), the benchmark questions, the retrieval depth ($k = 5$), and the generation temperature (0.0). In the generation stage (RQ3, RQ4), the retriever was fixed to Hybrid BM25–E5 at $k = 5$, and the generation model, prompt, and decoding settings were identical for both question sets (Table 12). No model was trained or fine-tuned, so no training and test partition applies. Separation was maintained between the 10 pilot questions and the final evaluation sets, and between the answerability labels and the generated outputs (Section 9.4).

Retrieval was evaluated on the full 135-question candidate benchmark ($N = 134$ for Recall@5). Generation and abstention were evaluated on a separate balanced benchmark of 60 Answerable and 60 Unanswerable questions (120 in total) that was drawn from the same 135-question pool after annotation.

3.7.2 Benchmark question collection

Legal research questions were collected from practicing judicial officers with a structured Google Form, in preference to crowd-sourced or synthetic queries. Each respondent recorded a contributor code or initials, a self-reported professional-experience band, a consent response, and five question slots. The raw response file had 30 rows. An embedded consent statement told respondents that their questions could be edited for grammar, formatting and consistency without changing their legal meaning, and could be annotated, categorized and included in an anonymized evaluation benchmark, with their identity withheld from the manuscript, related publications and the benchmark dataset. Twenty-nine respondents consented and one declined; the row of the non-consenting respondent was removed. Each of the 29 consenting respondents contributed five questions, giving 145 questions. Contributor codes were replaced by sequential identifiers JO01 to JO29, assigned by a sorted-name-to-index mapping. Self-reported experience was 11 to 20 years for 15 respondents, less than 5 years for 10, more than 20 years for 3 and 5 to 10 years for 1. No other respondent characteristics were recorded. The raw file also contained the placeholder codes TEST01 and TEST02. TEST02 was the row of the non-consenting respondent and was removed. TEST01 appeared on consenting rows and was not separately distinguished or investigated.

An exact-string duplicate check on the 145 questions found no duplicates; semantic or near-duplicate detection was not performed. Ten questions were then selected manually for a pilot, chosen to span criminal law, civil procedure, property law, arbitration, commercial law and special statutes, and were excluded from the candidate pool so that the final evaluation stayed independent of the pilot. The remaining 135 questions were randomized with a fixed seed (42) into a review order for annotation.

3.7.3 Relevance annotation

Annotation was applied to individual chunks. For each of the 135 candidate questions and for each of the six retrieval models, the top five retrieved chunks (k = 5) were annotated, giving 135 × 5 = 675 rows per model. Each chunk received one of three labels with a numeric level: Highly Relevant (2), Partially Relevant (1) or Not Relevant (0). A free-text comments field accompanied each row. A question-level "Uncertain" category, present in an early annotation template, was dropped, leaving a strict binary Answerable/Unanswerable scheme. Each completed annotation file passed an automated validation step that checked that labels belonged to the three valid categories, that numeric levels lay in 0, 1, 2, that labels and levels agreed, and that free text had not been entered in the answerability column. The same step corrected common typographical errors (for example, "artially Relevant" to "Partially Relevant" and "Not Answerable" to "Unanswerable"). This validation was adopted after the pilot revealed labeling irregularities (Section 9.5).

Annotation followed a single-annotator or limited-annotator protocol using structured rubrics and category definitions. Automated checks confirmed the internal consistency of each annotator's labels, but no second annotator cross-checked the labels, and no inter-rater agreement statistic (such as Cohen's kappa or Krippendorff's alpha) was computed for any annotation stage, including the relevance labels, the answerability adjudication, the answer-quality rubric and the abstention safety labels.

Pooled relevance set for Recall@5. Recall@5 requires a set of all relevant chunks per question, which was not available from an exhaustive corpus-wide judgment. It was therefore built by pooling: for each question, the relevant set was the union of the chunks judged relevant among the top-5 outputs of all six systems. The pool combined chunks with consistent binary labels across the six systems' annotation records, chunks with disagreeing labels that were resolved by majority vote, and 106 chunks with tied or unresolved disagreement that were adjudicated manually. It contained 2,870 unique question–chunk pairs, of which 1,505 were relevant and 1,365 were not relevant. The number of relevant chunks per question averaged 11.15 (median 12; range 0 to 24). One question (RAW118) had no relevant chunk in the pool, which left its Recall denominator undefined, and it was excluded from the Recall@5 averages, so that Recall@5 was computed for N = 134 questions and all other metrics for N = 135. Because the pool is limited to what the six systems retrieved, Recall@5 measures comparative coverage among these systems and does not estimate recall against all relevant material in the corpus.

3.7.4 Answerability labels and evaluation sets

Definitions. A question was labeled Answerable when the corpus contained sufficient legal evidence to support an evidence-grounded answer, and Unanswerable when the required legal information was absent from the corpus so that the system was expected to abstain. Operationally, these labels were derived from the chunk annotations by one fixed rule: a question is Answerable if at least one of its top-5 chunks was annotated Highly Relevant, and Unanswerable otherwise. The rule was applied to the annotation

of every retrieval model, and it therefore defines answerability relative to the top-5 chunks returned by the model concerned.

Carried-forward annotation. The Hybrid BM25–E5 annotation was adopted for the generation and abstention evaluation, because Hybrid was the best-performing retriever in the RQ2 analysis. Under this annotation, 73 of the 135 questions were Answerable and 62 were Unanswerable. An initial validation pass found that 48 Hybrid questions had non-unanimous question-level labels across their five annotated rows. Forty-five of these contained at least one Highly Relevant chunk and were set to Answerable by the rule. The remaining three (RAW107, RAW116, and RAW120) had no Highly Relevant chunk and were adjudicated manually, with a written rationale for each. In RAW107, the retrieved chunks stated general principles on the requirement of judicial reasoning but none specific to the Section 175(3) BNSS issue asked. In RAW116, the chunks addressed admissions and denials under the Code of Civil Procedure and Section 151 CPC in general, but not the question of admitting or denying a signature. In RAW120, the chunks covered arrest, detention, and remand in general, but not the remedy sought for an expired remand period. In each case, the same criterion applied: evidence that addressed the general legal area without resolving the specific issue asked did not meet the Highly Relevant threshold, and the question was adjudicated Unanswerable. After adjudication, every Hybrid question carried a single consistent label.

Sampling. A random sample of 60 of the 73 Answerable questions was drawn for RQ3, and a separate random sample of 60 of the 62 Unanswerable questions for RQ4, both with seed 42. This produced a balanced 120-question benchmark, which replaced an initially planned split of 50 Answerable and 25 Unanswerable questions and was chosen to give a cleaner confusion-matrix interpretation.

Freezing. The RQ3 and RQ4 question sets and their Hybrid top-5 evidence were frozen before generation (the RQ3 evidence file comprised 300 rows, five per question). The Unanswerable and Answerable ground-truth labels came only from the chunk-relevance annotation, were fixed before any model output was generated, and were not revisited afterwards. A targeted review of the workflow confirmed that no label was modified after generation.

3.7.5 Pilot studies

Two pilot studies with the same 10 questions (RAW001, RAW002, RAW004, RAW005, RAW006, RAW007, RAW010, RAW012, RAW015, and RAW017) validated the pipeline before the full-scale evaluation. They were treated as pipeline validation and not as confirmatory evidence, given $n = 10$.

In the retrieval pilot, the six configurations were run at $k = 5$ over the same chunk corpus, and Precision@5, Hit Rate@5, MRR, and nDCG@5 were computed. Recall@5 was deferred because the pooled relevance set did not yet exist. The pilot also ran underpowered versions of the RQ1 and RQ2 analyses (a one-sided Wilcoxon test on lexical and semantic family scores with `zero_method = "wilcox"`, and a Friedman test with Holm-corrected pairwise tests, 15 comparisons per metric).

In the generation pilot, Hybrid top-5 evidence was passed to Gemini `gemini-3.5-flash` at temperature 0.0, and the answers were rated on four dimensions (correctness, faithfulness, citation accuracy, and completeness). A `Qwen/Qwen3-4B-Instruct-2507` model was also loaded in 4-bit quantization on a Tesla T4 GPU but was not used to generate any reported output.

The pilots led to the following changes in the protocol.

- **Annotation irregularities.** The pilot annotation files contained a label with a stray double space, out-of-scale relevance values (3, 4, and 5) with bare-digit level codes, and a non-standard label ("Relevant"). These prompted the automated label validation described in Section 9.3.
- **Truncated generations.** With `max_output_tokens` = 1,200 and the default thinking-token budget, 5 of 10 answers were truncated mid-sentence, because the model's internal reasoning tokens (up to 1,149 on one question) consumed the output allowance. The pilot re-run used `thinking_level` = `"minimal"` and `max_output_tokens` = 3,000 with temperature 0.0.
- **Abstention detection.** An exact-string abstention detector failed to classify one answer that abstained in substance but added an explanatory sentence (RAW005). No code-level fix was documented, and string-based detection was carried into the full-scale classification (Section 9.7).
- **Post hoc relabeling.** Pilot ground-truth labels had been assigned before generation (8 Answerable, 2 Insufficient Evidence). After the first generation run, the label of RAW015 was changed from Answerable to Insufficient Evidence following an evidence review that the model's abstention had prompted. Because this revision followed observation of model output, the pilot abstention outcome is illustrative only, and no hypothesis test was run on it. This experience led to the rule, applied in the final evaluation, that ground-truth labels are derived from chunk annotation and frozen before any generation (Section 9.4).
- **Recall.** The pilot confirmed the need for Recall@5, which was introduced at full scale through the pooled relevance set (Section 9.3).

3.7.6 Answer-quality rubric (RQ3)

The 60 answers generated for the Answerable set were scored manually on five dimensions (Table 13), including answers that consisted of the abstention sentence. The rubric emphasizes faithfulness and relevance in the manner of automated RAG evaluation frameworks (Es et al., 2024) and relies on expert judgment of generated answers, as recommended for Indian legal question answering (Nigam et al., 2023). Four dimensions were scored 0 to 2. Citation Accuracy was also scored 0 to 2 but allowed a rating of Not Applicable, used when no citation was reasonably required or available; such a rating removed the dimension from the denominator for that answer instead of counting as zero. Two further fields were recorded but did not enter the composite score: `Case_Facts_Requested` (Yes or No, flagging questions that sought case facts rather than a legal proposition) and free-text evaluation comments.

Table 13 Answer-quality rubric.

Dimension	**2**	**1**	**0**
Legal correctness	Correct	Partially correct (material omission, qualification, or limited inaccuracy)	Incorrect (materially mis dicts the evidence)
Evidence grounding (faithfulness)	Fully grounded	Partially grounded (some unsupported inference or overstatement)	Ungrounded (fabricated, inconsistent)
Citation accuracy (or NA)	Accurate	Partially accurate	Inaccurate or fabricated
Answer relevance	Highly relevant	Partially relevant	Not relevant
Completeness	Complete	Partially complete	Incomplete (also covers v questions received the material)

3.7.7 Abstention annotation and behaviour classification (RQ4)

Two classification layers were applied to the combined 120-question set. For the 60 Unanswerable questions, each generated answer was annotated manually with five fields: `Model_Behaviour` (Correct Abstention or Unsafe Answer), `Safe_Abstention` (Yes/No), `Unsupported_Content` (Yes/No), `Hallucination` (Yes/No), and free-text safety comments. The decision rules were as follows. Correct Abstention: the evidence is insufficient and the model refuses or gives only a bounded statement of insufficiency. Unsafe Answer: the model gives a substantive legal answer that the frozen retrieved evidence does not support. Hallucination = Yes: the response asserts unsupported legal propositions, authorities, facts, statutes, holdings, or conclusions. A third category, Partial/Ambiguous, was defined for responses that combine appropriate qualification with potentially unsupported substantive content and therefore need adjudication. Before analysis, the annotation file was validated: all 60 frozen question identifiers matched with no missing, extra, or duplicate rows, `Model_Behaviour` values fell within the valid set, and logical-consistency assertions held (Correct Abstention implies `Safe_Abstention` = Yes and `Unsupported_Content` $\neq$ Yes; Unsafe Answer implies `Safe_Abstention` = No; Hallucination = Yes implies `Unsupported_Content` = Yes).

For the 60 Answerable questions, behaviour was classified automatically as Answered or Abstained by detecting the fixed abstention sentence in the generated text (substring matching). This classification was not verified by an independent manual re-annotation.

The 120 questions were combined into a confusion matrix in which Answerable ground truth and an Answered response constitute the positive class: true positives are Answerable questions that were answered, false negatives are Answerable questions on which the model abstained, false positives are Unanswerable questions that were answered, and true negatives are Unanswerable questions on which the model abstained.

3.8 Research questions and hypotheses

Four research questions guided the evaluation (Table 14). Each was paired with a null and an alternative hypothesis and a test specified in advance.

Table 14 Research questions, hypotheses and tests.

RQ	Question	H_0	H_1	Test
RQ1	Do semantic retrieval models significantly outperform lexical retrieval models for Indian Supreme Court legal question answering?	Semantic (E5, BGE, InLegalBERT) and lexical (TF-IDF, BM25) retrieval perform equally.	Semantic retrieval performs significantly better than lexical retrieval.	One-sided Wi (paired, per qu
RQ2	Which single retrieval model performs best overall among TF-IDF, BM25, E5, BGE, InLegalBERT and the Hybrid BM25–E5 retriever?	The six methods do not differ in performance.	At least one method differs significantly from the others.	Friedman omni corrected pairw where significa
RQ3	How accurately does the framework generate evidence-grounded legal answers based on retrieved Supreme Court judgments for the year 2024?	The median answer-quality score is not significantly greater than the benchmark score.	The median answer-quality score is significantly greater than the benchmark score.	One-sided on signed-rank
RQ4	Can the framework safely abstain from answering when the retrieved Supreme Court evidence is insufficient to support a reliable legal answer?	The true correct-abstention proportion is at or below the benchmark proportion.	The true correct-abstention proportion is significantly greater than the benchmark proportion.	Exact one-sided

Before the retrieval evaluation, directional expectations were also stated for the three dense models (Table 15). They concern the direction of differences and do not predict metric values, because no earlier study had benchmarked these models on Indian Supreme Court retrieval. In particular, whether InLegalBERT's domain-vocabulary advantage would outweigh its lack of a retrieval-specific training objective was treated as an open empirical question.

Table 15 Directional expectations for the dense embedding models.

Model	Training approach	Expected behaviour	Rationale
E5	Contrastive pre-training on large-scale general-domain text pairs with a "query: " / "passage: " prefix convention (Wang et al., 2022)	Performs well on general semantic similarity matching, comparable to or exceeding other general-purpose dense embeddings, with no legal-domain adaptation	The contrastive objective is designed to optimize e ding geometry for retrieval-style similarity; the prefixes retrieval intent but do not encode legal vocabulary or c structure
BGE	General-purpose packaged embedding resource trained on broad cross-domain corpora, without prefixes (Xiao et al., 2024)	Comparable to, or somewhat different from, E5, with strong general cross-domain retrieval orientation and no legal specialization	E5 and BGE pursue similar goals through different tr pipelines and data mixtures; without legal-domain trainin ferences were expected to be modest
InLegalBERT	Domain-adaptive pre-training of a BERT-family encoder on Indian legal text (Paul et al., 2023)	Domain-vocabulary advantage for Indian legal terminology and citation patterns, but flagged as a possible limitation relative to retrieval-specific models	Legal-domain pre-training should improve the represen of statute names, citation formats and legal phrasin the embedding space may be less suited to nearest-neig retrieval because the model is a masked-language-model e without a retrieval or contrastive objective

3.9 Evaluation metrics

3.9.1 Retrieval

All retrieval metrics were computed at $k = 5$ for each question and then averaged over questions. Relevance judgments came from the chunk-level annotation of Section 9.3; retrieval-stage metrics of this kind rest on established relevance-judgment methodology (Al-Maskari et al., 2008). For the binary metrics, a chunk was treated as relevant when its annotated level exceeded 0 (Partially or Highly Relevant).

- **Precision@5:** the proportion of the five retrieved chunks judged relevant.
- **Recall@5:** the number of relevant chunks retrieved divided by the number of relevant chunks in the pooled relevance set ($N = 134$; Section 9.3).
- **Hit Rate@5:** 1 when at least one relevant chunk appears in the top five, and 0 otherwise.
- **Mean reciprocal rank (MRR):** the mean over questions of the reciprocal of the rank of the first relevant chunk, with 0 when none is retrieved.
- **nDCG@5:** normalized discounted cumulative gain over the five ranked chunks.

Precision@5, Hit Rate@5, MRR, and nDCG@5 were averaged over $N = 135$ questions. Means were reported without standard deviations or confidence intervals; statistical uncertainty is conveyed through the hypothesis tests of Section 12.

3.9.2 Answer quality

The RQ3 outcome is the **Composite Quality Score** of each answer:

$$\text{Composite Score} = \frac{\sum \text{applicable dimension scores}}{2 \times \text{number of applicable dimensions}}, \tag{2}$$

which lies between 0 and 1. It is equivalent to the mean rubric score across the applicable dimensions divided by 2. The predefined acceptability benchmark is **0.75**, corresponding to an average rubric score of 1.5 out of 2 across applicable dimensions.

For descriptive reporting, composite scores were grouped into four bands: **Very High** (≥ 0.90), **High** (0.75–0.89), **Moderate** (0.50–0.74), and **Low** (< 0.50). Per-dimension means, medians, standard deviations, and ranges were also computed.

3.9.3 Abstention and safety

From the combined confusion matrix (Section 9.7), the following metrics were computed with *Answerable* and *Answered* as the positive class: accuracy, precision, recall, F1-score, specificity, and false-positive rate. Precision, recall, specificity, and false-positive rate were calculated as:

$$\text{Precision} = \frac{TP}{TP + FP}, \tag{3}$$

$$\text{Recall} = \frac{TP}{TP + FN}, \tag{4}$$

$$\text{Specificity} = \frac{TN}{TN + FP}, \tag{5}$$

$$\text{False Positive Rate} = \frac{FP}{FP + TN}. \tag{6}$$

Specificity is equivalently the **safe-abstention rate**, whereas the false-positive rate is equivalently the **unsafe-answer rate**. Within the 60 Unanswerable questions, the safe-abstention rate therefore equals specificity, and the unsafe-answer rate equals the false-positive rate.

The manual safety annotation additionally provided the **unsupported-content rate** and **hallucination rate** among the 60 Unanswerable questions. The inferential outcome for RQ4 was the **number of correct abstentions** among these 60 Unanswerable questions.

3.10 Statistical analysis

Non-parametric tests were used because they do not assume normally distributed paired differences, in line with guidance for comparing repeated-measures performance of algorithms (Wilcoxon, 1945; Derrac et al., 2011). The significance level was $\alpha = 0.05$ throughout. The unit of analysis was the question.

RQ1.

For each of the 135 questions, a lexical family score (the mean of the TF-IDF and BM25 scores) and a semantic family score (the mean of the E5, BGE and InLegal-BERT scores) were calculated separately for Precision@5, MRR and nDCG@5, with the Hybrid model excluded from both families. The paired difference was defined as semantic minus lexical, so that a positive value favours semantic retrieval. A one-sided Wilcoxon signed-rank test (H_1: semantic > lexical) was run for each of the three metrics on $N = 135$ pairs, with `zero_method="wilcox"`, which discards pairs with an exactly zero difference before ranking. Hit Rate@5 was summarized descriptively at family level only and was not tested, because it is a binary per-question indicator, which does not suit the continuous, symmetric difference distribution assumed by the Wilcoxon test.

The effect size was the rank-biserial correlation of the semantic-minus-lexical differences, computed after removing zero differences, and it was classified as negligible ($|r| < 0.10$), small (0.10–0.29), moderate (0.30–0.49), or large (0.50 or above). The three RQ1 tests were not corrected for multiple comparisons.

RQ2.

A descriptive mean rank was first computed by ranking the six models on each of the five metrics and averaging the ranks per model; this was calculated independently of, and before, the Friedman test. Friedman tests (six related samples matched by question; H_0: the six models share the same performance distribution) were then run separately for Precision@5 ($N = 135$), Recall@5 ($N = 134$), MRR ($N = 135$), and nDCG@5 ($N = 135$).

Post-hoc pairwise Wilcoxon signed-rank tests were specified for metrics with a significant omnibus test and were carried out for all 15 model pairs on each of the four metrics (60 tests in total), with Holm's sequentially rejective correction (Holm, 1979).

Effect sizes for the pairwise comparisons were rank-biserial correlations, interpreted with the thresholds above. Kendall's W (Kendall & Babington Smith, 1939) had been planned as an additional effect-size measure for the Friedman test but was not computed in the final analysis. RQ2 is therefore based on the Friedman χ^2, its degrees of freedom and p-value, the Holm-corrected pairwise tests, and the rank-biserial effect sizes.

RQ3.

The 60 composite scores were compared with the predefined benchmark of 0.75 by a one-sided one-sample Wilcoxon signed-rank test (H_0: median composite score ≤ 0.75; H_1: median > 0.75; $\alpha = 0.05$), reporting an exact p-value. The effect size was the matched-pairs rank-biserial correlation,

$$r = \frac{W^+ - W^-}{W^+ + W^-}, \tag{7}$$

where W^+ and W^- are the sums of positive and negative signed ranks relative to the benchmark. The effect size was interpreted using the thresholds defined above.

RQ4.

The number of correct abstentions among the 60 Unanswerable questions was tested against a predefined benchmark proportion of 0.80 with an exact one-sided binomial test (H_0: rate ≤ 0.80; H_1: rate > 0.80; $\alpha = 0.05$), which compares an observed proportion with a fixed benchmark without a normal approximation (Laurencelle, 2021). The benchmark corresponds to safely abstaining on at least four of every five evidence-insufficient queries. A 95% exact confidence interval was computed.

The effect size was Cohen's h,

$$h = 2 \arcsin\left(\sqrt{p_1}\right) - 2 \arcsin\left(\sqrt{p_0}\right), \tag{8}$$

where p_1 is the observed proportion and p_0 is the benchmark. Cohen's h was classified as negligible (< 0.20), small (< 0.50), medium (< 0.80), or large (≥ 0.80).

The sample sizes entering the tests were $N = 135$ for RQ1 and for the RQ2 tests on Precision@5, MRR, and nDCG@5; $N = 134$ for the RQ2 tests on Recall@5; $N = 60$ for RQ3; and $N = 60$ for RQ4.

3.11 Sample-size and power analysis

Sample sizes were determined prospectively with $\alpha = 0.05$ and a target power of 0.80. The planning framework specified two-sided tests unless a directional test was justified; one-sided tests were adopted for RQ1, RQ3, and RQ4 because their hypotheses are directional, whereas the Friedman test for RQ2 is omnibus and non-directional. Closed-form power formulas are not available for the Wilcoxon signed-rank and Friedman tests, so power was approximated for planning purposes. For RQ1 and RQ3, the approximation was the paired or one-sample normal (t-test) power calculation using `statsmodels TTestPower`. For RQ2, an asymptotic non-central chi-square approximation based on Kendall's W (Kendall & Babington Smith, 1939) was used, with non-centrality

$$\lambda = N(k-1)W, \tag{9}$$

where $k = 6$ models, 5 degrees of freedom, and a critical χ^2 value of 11.070. For RQ4, the exact binomial distribution using `scipy.stats.binom` allowed an exact power calculation. In every case, the hypothesis tests applied to the final data were those specified in Section 12. Candidate sample sizes from 5 to 100 were examined for each analysis.

No pilot-derived effect-size estimate was available at the design stage. Therefore, each effect size was adopted as a conservative prospective assumption and was not taken from a cited source: a standardized paired effect $d_z = 0.50$ for RQ1, a standardized one-sample effect $d = 0.50$ for RQ3, and Kendall's $W = 0.20$ for RQ2, corresponding to conventional medium-effect benchmarks. For RQ4, the anticipated correct-abstention proportion under the alternative hypothesis was $p_1 = 0.85$, representing the expected performance of the system, and a minimum-acceptable benchmark p_0 was selected after comparison of three candidate values (Table 17). The resulting minimum sample sizes and the sample sizes used are shown in Table 16.

Table 16 Prospective minimum sample sizes and the sample sizes used.

RQ	Test	Assumed effect size	Minimum N	Power at minimum N	Sample size in the original plan	Sample s
RQ1	One-sided Wilcoxon signed-rank (normal approximation)	$d_z = 0.50$	27 answerable questions (analytical estimate 26.138)	0.812 (0.798 at $N = 26$)	50 (power 0.967)	135 quest
RQ2	Friedman test (chi-square approximation)	$W = 0.20$	13 answerable questions	0.806 (0.769 at $N = 12$)	50 (power 1.000)	135 quest Recall@5
RQ3	One-sample Wilcoxon signed-rank (normal approximation)	$d = 0.50$	27 answerable questions (analytical estimate 26.138)	0.812	50 (power 0.967)	60 Answ tions
RQ4	Exact one-sided binomial test ($p_0 = 0.60$, $p_1 = 0.85$)	$p_1 = 0.85$	21 Unanswerable questions (17 correct abstentions needed to reject H_0)	0.803	25	60 Unans tions

RQ1 and RQ3 share the same approximation and effect size, so their power curves are numerically identical.

Table 17 Candidate benchmark proportions considered for the RQ4 sample-size analysis ($p_1 = 0.85$).

Benchmark p_0	Minimum N	Critical correct abstentions	Estimated power	Stated rationale
0.50	13	10	0.882	Tests performance above a basic majority-level threshold; smallest sample requirement but a comparatively weak standard
0.60	21	17	0.803	Moderate minimum acceptable level; intermediate and practically feasible threshold that still requires performance clearly above a majority-level benchmark
0.70	49	40	0.809	Comparatively demanding, safety-oriented standard; substantially larger sample required

The value $p_0 = 0.60$ was adopted for sample-size planning. The originally planned benchmark comprised 50 Answerable questions for RQ1–RQ3 and 25 Unanswerable questions for RQ4 (75 in total). The final design differed from this plan in three respects: retrieval evaluation used all 135 candidate questions in place of 50; the generation and abstention evaluation used a balanced 60/60 design in place of 50/25; and Kendall's W, planned for RQ2, was not computed. Every executed sample size exceeded its prospective minimum (27 for RQ1 and RQ3, 13 for RQ2, and 21 for RQ4). In addition, the confirmatory RQ4 test used a benchmark proportion of 0.80, which differs from the $p_0 = 0.60$ assumed for sample-size planning.

3.11.1 Software, computing environment and availability

Analyses were run in Google Colab with Python 3.12.13. Recorded library versions were Pandas 2.2.2, NumPy 2.0.2, Matplotlib 3.10.0, PyMuPDF 1.28.0, and ChromaDB 1.5.9, while the deployment interface used Gradio 6.20.0. The lexical and dense retrieval components used `rank_bm25`, scikit-learn (with joblib and `scipy.sparse` for persisting the TF-IDF artifacts), and sentence-transformers; their versions were not recorded. Power calculations used `statsmodels` and `scipy`. Generation used the `google.genai` SDK with `gemini-3.6-flash`. Device selection for embedding was left to the automatic behaviour of sentence-transformers, and GPU use was not confirmed. A fixed seed of 42 was used for the Python and NumPy random generators during data preparation and for the sampling of the RQ3 and RQ4 evaluation sets.

The pipeline was organized as six sequential notebooks. The first performs acquisition, cleaning, exploratory analysis, chunking, and index construction. The second carries out the prospective power analysis. The third builds the benchmark question set. The fourth and fifth run the retrieval and the generation and abstention pilots, respectively. The sixth runs the full retrieval evaluation, the annotation workflow, the RQ1–RQ4 analyses, construction of the frozen RQ3 and RQ4 sets, and the Gradio deployment.

Key artifacts include the chunk corpus (`chunks_384_40.parquet`), the dense embedding matrices (30,521 $\times$ 768 for each dense model), the persistent ChromaDB collections, the persisted TF-IDF and BM25 indices, the 135-question candidate benchmark, the per-model relevance annotation files, the frozen RQ3 and RQ4 evaluation sets with their evidence, the generated answers, and the rubric and safety annotation files. Earlier superseded versions of some artifacts, including files from a previous 19,237-chunk configuration, are stored in an archive folder and are not part of the pipeline that yields the reported results.

Reproduction requires the 782-judgment corpus from the AWS Open Data registry, a Google Gemini API key provisioned for `gemini-3.6-flash`, and the environment described above. The project repository is available at https://github.com/harshalichowdhary-design/NyayaRAG.

4 Results

The evaluation was conducted in two distinct stages. Retrieval performance was assessed using the complete candidate benchmark of 135 questions, with 134 questions

contributing to Recall@5 because one question had an undefined recall denominator. Generation quality and abstention were evaluated separately on a balanced benchmark comprising 60 Answerable and 60 Unanswerable questions. These populations should therefore be interpreted separately rather than as a single evaluation set.

4.1 Pilot Study Results

Two pilot evaluations, each containing 10 questions, were conducted before the full-scale experiments. The first evaluated the six retrieval configurations, whereas the second examined grounded answer generation and abstention. The pilot questions were excluded from the final benchmark, and the pilot findings were treated as pipeline-validation evidence rather than confirmatory results because of the small sample size.

In the retrieval pilot, all six configurations were evaluated at $k = 5$. The Hybrid BM25–E5 configuration obtained a Precision@5 of 0.74 and nDCG@5 of 0.8088, while E5 produced the highest MRR of 0.85 and tied the Hybrid configuration on Hit Rate@5 at 0.90. InLegalBERT recorded the lowest results across the evaluated retrieval measures. The exploratory RQ1 comparison did not produce a statistically significant result, with the smallest observed p-value being 0.375. A pilot Friedman analysis for RQ2 identified significance for Precision@5 and MRR; however, none of the 15 pairwise comparisons remained significant following Holm correction. The pilot also confirmed that the six retrieval pipelines and annotation/metric workflow could be executed end to end. It additionally identified the need to incorporate Recall@5 into the full-scale evaluation through a pooled-relevance procedure.

The generation and abstention pilot used the Hybrid retriever with Gemini (`gemini-3.5-flash`) at temperature 0.0. Seven of the 10 questions received substantive answers, while three resulted in abstention. Five of the seven generated answers were scored because two were excluded for parser/annotation reasons; all five scored answers were judged fully correct, grounded, cited, and complete. No composite score or hypothesis test was calculated for this pilot.

An important methodological issue was identified during the pilot. Although the initial ground-truth labels comprised eight Answerable and two Insufficient Evidence questions, one question (RAW015) was subsequently relabeled as Insufficient Evidence after the evaluator reviewed the evidence following the model's abstention. Consequently, the pilot confusion matrix, despite producing correct decisions for all 10 questions, was not treated as confirmatory evidence. The final 60/60 benchmark avoided this procedure by fixing answerability labels from evidence-based relevance annotation before generation.

The pilot also identified two implementation issues. A 1,200-token output limit resulted in five of ten generated answers being truncated mid-sentence; this was addressed by setting `thinking_level = "minimal"` and retaining the required token allowance. In addition, an exact-string abstention detector failed to recognize one substantively abstaining response that contained explanatory text. This limitation remained relevant to the subsequent classification procedure.

4.2 Relevance Annotation and Benchmark Finalization

The 135 candidate questions were manually annotated against the top five retrieved chunks from each of the six retrieval configurations. Each chunk received a relevance label of Highly Relevant, Partially Relevant, or Not Relevant, corresponding to scores of 2, 1, and 0, respectively. Question-level answerability was then derived using a fixed rule: a question was considered Answerable when at least one of its top-five retrieved chunks was Highly Relevant; otherwise, it was classified as Unanswerable.

The resulting answerability counts varied across retrieval configurations. TF-IDF yielded 51 Answerable and 84 Unanswerable questions; BM25, 67 and 68; E5, 67 and 68; BGE, 60 and 75; and InLegalBERT, 40 and 95, respectively. The Hybrid BM25–E5 configuration produced 73 Answerable and 62 Unanswerable questions.

Because the subsequent retrieval comparison established the Hybrid configuration as the retrieval configuration carried forward into generation, its relevance annotations were used for the final benchmark construction. Among the 135 Hybrid-labelled questions, 48 initially received non-unanimous answerability labels. Forty-five were resolved automatically as Answerable, while three questions (RAW107, RAW116, and RAW120) were manually adjudicated as Unanswerable because the retrieved evidence addressed the general legal subject but not the particular fact pattern.

Using a reproducible random selection procedure with seed 42, 60 of the 73 Answerable questions were sampled for RQ3 and 60 of the 62 Unanswerable questions were sampled for RQ4. This produced the final balanced benchmark of 120 questions and exceeded the prospective minimum sample requirements of 27 for RQ3 and 21 for RQ4.

4.3 Final Retrieval Results: RQ1 and RQ2

4.3.1 Retrieval Performance Across Models

All six retrieval configurations were evaluated over the complete 135-question candidate benchmark, except for Recall@5, which used 134 questions. The measured retrieval performance is shown in Table 18.

Table 18 Final retrieval performance across the six configurations

Retrieval model	Precision@5	Recall@5	Hit Rate@5	MRR	nDCG@5
TF-IDF	0.5704	0.2674	0.7481	0.6585	0.6722
BM25	0.6030	0.2848	0.8296	0.6977	0.7113
E5	0.6311	0.2831	0.8593	0.7621	0.7396
BGE	0.5985	0.2723	0.8444	0.7494	0.7396
InLegalBERT	0.4637	0.1918	0.7037	0.5958	0.5979
Hybrid BM25–E5	0.8089	0.3318	0.9778	0.9184	0.8839

The Hybrid configuration recorded the highest descriptive value for each reported retrieval metric. E5 and BGE produced similar values on several measures, while InLegalBERT recorded the lowest value for each metric. Recall@5 was calculated against a pooled relevance set formed from the union of relevant chunks identified across the

six systems; therefore, this metric represents comparative retrieval coverage within the evaluated systems rather than exhaustive recall over the entire judgment corpus.

4.3.2 RQ1: Semantic Versus Lexical Retrieval

RQ1 tested whether semantic retrieval provided statistically higher performance than lexical retrieval. For each question, the lexical-family score was calculated as the mean of TF-IDF and BM25, whereas the semantic-family score was calculated from E5, BGE, and InLegalBERT. The Hybrid configuration was excluded from both groups.

A one-sided paired Wilcoxon signed-rank test was applied to Precision@5, MRR, and nDCG@5. Precision@5 produced lexical and semantic means of 0.5867 and 0.5644, respectively, with $W = 3747.0$, $p = 0.8176$, and rank-biserial $r = -0.0923$. For MRR, the corresponding means were 0.6781 and 0.7024, with $W = 2741.5$, $p = 0.2868$, and $r = 0.0645$. For nDCG@5, the means were 0.6917 and 0.6924, with $W = 4249.0$, $p = 0.5675$, and $r = -0.0171$. Hit Rate@5 was reported descriptively but was not subjected to the Wilcoxon test.

None of the three inferential comparisons reached the predefined significance criterion. Moreover, all three effect sizes were negligible. Thus, the observed data did not provide statistical evidence that the semantic retrieval family outperformed the lexical family on this benchmark.

4.3.3 RQ2: Comparison of the Six Retrieval Configurations

RQ2 evaluated whether retrieval performance differed among the six configurations. The Friedman omnibus test was significant for all four inferential retrieval metrics. For Precision@5, $\chi^2(5) = 107.6769$, $p < .001$; for Recall@5, $\chi^2(5) = 52.5064$, $p < .001$; for MRR, $\chi^2(5) = 69.4665$, $p < .001$; and for nDCG@5, $\chi^2(5) = 37.6621$, $p < .001$. The null hypothesis of identical performance distributions was therefore rejected for each metric.

Holm-corrected pairwise Wilcoxon comparisons produced 60 comparisons in total, of which 30 were statistically significant. The Hybrid configuration was significantly different from every individual retriever for Precision@5, MRR, and nDCG@5, and from four of the five individual models for Recall@5. The BM25 comparison for Recall@5 was the exception ($p = 0.1163$). Among the 30 significant comparisons, 19 had large rank-biserial effects and 11 had moderate effects; none were classified as negligible or small.

The Hybrid configuration was consequently selected for the downstream generation and abstention experiments. This selection was based on its descriptive retrieval performance, the significant omnibus and pairwise comparisons, and the associated effect sizes.

4.3.4 Hybrid Retrieval Composition

The Hybrid retriever combined BM25 and E5 using unweighted Reciprocal Rank Fusion:

$$RRF(d) = \frac{1}{k + r_{BM25}(d)} + \frac{1}{k + r_{E5}(d)}, \qquad k = 60. \tag{10}$$

Each component supplied its top 20 candidates, after which the candidates were merged and ordered according to their RRF scores before retaining the final five chunks.

Analysis of the 675 retrieved Hybrid chunk instances showed that 46.96% were retrieved by both BM25 and E5, 27.56% were surfaced only by BM25, and 25.48% only by E5. Thus, the final evidence set contained contributions from both retrieval mechanisms rather than being composed primarily of one component's results.

4.4 Generation Pipeline

Following the retrieval comparison, the Hybrid BM25–E5 configuration was fixed at a retrieval depth of five for RQ3 and RQ4. The generation model was Google's Gemini `gemini-3.6-flash`, with temperature 0.0, top-p 1.0, and a maximum output allowance of 1,200 tokens during the frozen evaluation.

The generation prompt supplied the question together with five evidence blocks containing the case title, judgment identifier, chunk identifier, and retrieved text. The model was instructed to construct the answer exclusively from the supplied evidence and not to introduce unsupported legal knowledge, case names, citations, or statutory provisions. When the evidence was judged insufficient, the prescribed abstention response was:

> The retrieved evidence is insufficient to provide a reliable answer.

No numerical confidence threshold was used for abstention; downstream classification instead detected this fixed abstention phrase.

4.5 Answer-Quality Results: RQ3

RQ3 evaluated the quality of evidence-grounded answers over the 60 Answerable questions. Each response was scored using five dimensions: Legal Correctness, Evidence Grounding/Faithfulness, Citation Accuracy, Answer Relevance, and Completeness. Each dimension used a 0–2 scale, except that Citation Accuracy could be marked not applicable. The composite score was calculated by dividing the sum of applicable dimension scores by twice the number of applicable dimensions, resulting in a normalized range from 0 to 1.

Table 19 RQ3 descriptive results

Dimension	N	Mean	Median	SD	Min–Max
Legal Correctness	60	1.7500	2.0	0.6277	0.0–2.0
Evidence Grounding	60	1.8833	2.0	0.3237	1.0–2.0
Citation Accuracy	46	2.0000	2.0	0.0000	2.0–2.0
Answer Relevance	60	1.8667	2.0	0.3428	1.0–2.0
Completeness	60	1.6500	2.0	0.6846	0.0–2.0
Composite Score	60	0.8983	1.0000	0.2334	0.25–1.00

Citation Accuracy was applicable to 46 of the 60 answers and was marked not applicable for the remaining 14. Forty-six answers (76.67%) achieved a perfect composite score of 1.00. Fifty answers (83.33%) fell within the Very High quality band (≥ 0.90), while two were classified as High, two as Moderate, and six as Low (< 0.50). Overall, 52 of the 60 answers (86.67%) reached or exceeded the predefined acceptability threshold of 0.75.

A one-sided one-sample Wilcoxon signed-rank test compared the composite scores with the 0.75 benchmark. The analysis yielded $W = 1452.0$, $p = 0.000013$, with rank-biserial $r = 0.5869$. The predefined statistical criterion was therefore met, with a large observed effect size.

Among the individual dimensions, Evidence Grounding had a mean of 1.8833 and Answer Relevance a mean of 1.8667. Citation Accuracy was perfect whenever applicable. Completeness had the lowest mean score at 1.6500 and represented the principal residual weakness identified in the evaluation. The documented examples included RAW003, which received a composite score of 0.90, and RAW022, which received 1.00. RAW019, although labelled Answerable, resulted in the fixed abstention response and received a composite score of 0.25.

4.6 Abstention and Safety Results: RQ4

RQ4 examined whether the system could abstain when the retrieved evidence was insufficient. The evaluation used the 60 Unanswerable questions from the finalized benchmark. Safety annotation was based on model behaviour, safe abstention, unsupported content, and hallucination, with internal consistency checks passing.

Of the 60 genuinely Unanswerable questions, 59 were correctly abstained on, corresponding to a safe-abstention rate of 98.33%. One question resulted in an unsupported answer, corresponding to an unsafe-answer rate of 1.67%; this was also the only case flagged as hallucinated.

The combined 120-question confusion matrix contained 46 true positives, 14 false negatives, one false positive, and 59 true negatives.

Table 20 Combined RQ3/RQ4 classification results

Metric	Value
Accuracy	0.8750 (87.50%)
Precision	0.9787 (97.87%)
Recall	0.7667 (76.67%)
F1-score	0.8598 (85.98%)
Specificity / Safe Abstention Rate	0.9833 (98.33%)
False Positive Rate / Unsafe Answer Rate	0.0167 (1.67%)

An exact one-sided binomial test compared the observed safe-abstention rate with the predefined benchmark of 0.80. With 59 successful abstentions out of 60 questions, the test produced $p = 0.000025$, with a 95% confidence interval of [0.9234, 1.0000]

and Cohen's $h = 0.6684$. The observed proportion therefore exceeded the specified benchmark under the prespecified statistical criterion.

The 14 false negatives in the combined matrix corresponded to Answerable questions on which the system abstained. Thus, within the evaluated benchmark, the more frequent classification error was over-abstention rather than unsafe answering. The single false positive represented the only unsafe answer among the 60 Unanswerable questions.

4.7 Consolidated Results

Table 21 Consolidated results by research question

Research question	Descriptive result	Statistical analysis	Result
RQ1: Lexical vs. semantic retrieval	No statistically significant semantic-family advantage on Precision@5, MRR, or nDCG@5	One-sided paired Wilcoxon signed-rank; $p = 0.8176$, 0.2868, and 0.5675	Null hypothesis not rejected
RQ2: Retrieval-model comparison	Hybrid BM25–E5 recorded the highest value on all five retrieval metrics	Friedman tests, all $p < .001$, followed by Holm-corrected pairwise tests	Null hypothesis rejected
RQ3: Evidence-grounded generation	Mean composite score = 0.8983; 86.67% reached $\geq$ 0.75	One-sample Wilcoxon: $W = 1452.0$, $p = 0.000013$, $r = 0.5869$	Benchmark exceeded
RQ4: Safe abstention	59/60 Unanswerable questions correctly abstained on (98.33%)	Exact binomial: $p = 0.000025$, 95% CI [0.9234, 1.0000], $h = 0.6684$	Benchmark exceeded

Taken together, the retrieval experiments showed that semantic retrieval did not achieve a statistically significant advantage over the lexical family when evaluated as groups. However, combining BM25 and E5 through Reciprocal Rank Fusion produced substantial improvements across the principal retrieval measures and provided the retrieval configuration used in the downstream generation experiments. The generation evaluation subsequently recorded a mean composite quality score of 0.8983, with 52 of 60 answers meeting the 0.75 acceptability threshold. The abstention evaluation recorded 59 correct abstentions among 60 Unanswerable questions, while the combined confusion matrix showed that over-abstention on Answerable questions was more frequent than unsafe answering.

4.8 Error Analysis

Five principal error or limitation patterns were documented.

First, the Hybrid evidence sets contained substantial complementary contributions from the two component retrievers. Across the 675 Hybrid-retrieved chunks, 27.56% were BM25-only, 25.48% were E5-only, and 46.96% were retrieved by both. RAW013

provides an illustrative case: none of its five Hybrid results were shared by both components, while two were BM25-only and three were E5-only.

Second, three questions—RAW107, RAW116, and RAW120—represented answerability edge cases. Their retrieved evidence addressed the general legal domain but did not establish the specific facts required by the questions. They were therefore adjudicated as Unanswerable.

Third, over-abstention constituted the principal observed classification error in the combined evaluation. Fourteen of the 60 Answerable questions were classified as abstentions despite having sufficient evidence, including RAW019.

Fourth, the citation format used during RQ3/RQ4 evaluation contained only `case_title` and `judgment_id`, rather than a formally formatted legal citation. A fuller citation field was subsequently constructed for the deployment interface. Consequently, the Citation Accuracy scores in the frozen evaluation refer to the simpler case-title/judgment-ID representation.

Finally, an inconsistency was observed in automated behavioural classification. An initial automated pre-screen classified the 60 RQ3 answers as 50 Answered and 10 Abstained, whereas the classification used for the final confusion matrix produced 46 Answered and 14 Abstained. Both classifications relied on substring matching rather than an independently documented manual re-verification. This limits the precision with which the exact false-negative count should be interpreted.

5 Discussion

The findings of this study provide a differentiated view of retrieval and generation performance in legal question answering over Indian Supreme Court judgments. Rather than showing a uniform advantage for one retrieval paradigm, the results indicate that lexical and semantic retrieval contributed complementary signals, with their combination producing the strongest retrieval performance under the evaluated conditions. The downstream generation experiments further showed that the selected retrieval configuration could support high-quality evidence-grounded answers while maintaining a high rate of abstention when the available evidence was insufficient. These findings should nevertheless be interpreted within the boundaries of the evaluated 2024 English-language Supreme Court corpus, benchmark, retrieval configurations, and generation pipeline. The discussion below therefore considers each research question separately before examining the broader methodological and practical implications.

5.1 RQ1: Semantic Versus Lexical Retrieval

RQ1 examined whether semantic retrieval models—E5, BGE, and InLegalBERT—would significantly outperform the lexical retrieval models TF-IDF and BM25. The results did not provide statistical evidence for such an advantage. The one-sided paired Wilcoxon comparisons produced p-values of 0.8176 for Precision@5, 0.2868 for MRR, and 0.5675 for nDCG@5, while the corresponding rank-biserial effect sizes were negligible. Thus, the predefined alternative hypothesis was not supported on the evaluated benchmark. Importantly, this result indicates insufficient evidence for semantic superiority rather than statistical equivalence between the two retrieval paradigms. The

distinction is relevant because a non-significant result does not establish that the retrieval families perform identically.

The RQ1 finding differs from results reported in adjacent retrieval and legal-QA literature. Karpukhin et al. (2020), for example, demonstrated advantages for dense passage retrieval over lexical retrieval in open-domain question answering, while Nigam et al. (2023) reported measurable benefits from semantic retrieval in an Indian legal setting. The present result therefore suggests that the conditions under which semantic retrieval gains an advantage are not necessarily reproduced in this particular Indian Supreme Court benchmark. At the same time, the finding is compatible with broader retrieval evidence indicating that no single retrieval paradigm dominates uniformly across domains (Thakur et al., 2021). The result consequently provides a more domain-specific qualification of the general expectation that semantic representations should automatically outperform lexical matching.

Several explanations documented in the study provide plausible interpretations of this null result. First, many judicial questions contain strong lexical signals, including statutory provisions, section numbers, case names, and established legal terminology. Such queries provide BM25 and TF-IDF with substantial term-overlap information, reducing the vocabulary-mismatch advantage that motivates dense retrieval. Second, the benchmark contains heterogeneous question types, meaning that some questions may benefit more from semantic matching while others favor lexical matching; opposing question-level effects can therefore be attenuated when aggregated at the retrieval-family level.

The composition of the semantic family is also relevant. E5 and BGE were individually stronger dense retrieval approaches, whereas InLegalBERT was comparatively weaker. Because the RQ1 semantic family was evaluated as an aggregate containing all three dense models, the lower performance of InLegalBERT contributed to the family-level result. This is particularly noteworthy given the preregistered expectation that InLegalBERT's Indian legal-domain pre-training could provide a vocabulary advantage. The results suggest that domain-specific language representation alone does not necessarily translate into superior nearest-neighbor retrieval when the encoder has not been trained specifically for retrieval.

The study also identified overlap between lexical and semantic retrieval outputs as another possible contributor. When both paradigms retrieve substantially similar relevant chunks, their aggregate performance can converge even if their underlying representation mechanisms differ. Furthermore, semantic retrieval showed a different strength profile across metrics: the semantic family performed comparatively better on ranking-oriented measures such as MRR while not establishing an advantage in the quantity of relevant material retrieved at the top ranks. The fixed 384-word chunking strategy may also have influenced this comparison because legal propositions can extend across paragraph or chunk boundaries, potentially affecting semantic similarity differently from lexical matching.

These observations are consistent with the benchmark characteristics. The questions were contributed by judicial officers and therefore reflect professional legal-research queries rather than artificially generated vocabulary-mismatch queries. Such questions may naturally contain terminology that overlaps with the judgment corpus.

Consequently, the absence of a significant semantic-family advantage should be interpreted as a property of the evaluated retrieval task rather than evidence that semantic retrieval lacks value in legal information retrieval. The result instead motivated the subsequent examination of whether the complementary behavior of the two paradigms could be exploited through hybrid retrieval. :chatgpt-content-referenceindex="1"

5.2 RQ2: Comparison of Retrieval Configurations

RQ2 examined whether retrieval performance differed among TF-IDF, BM25, E5, BGE, InLegalBERT, and the Hybrid BM25–E5 configuration. In contrast to RQ1, the six-way comparison produced statistically significant differences across all four inferential metrics. The Friedman tests were significant for Precision@5 ($\chi^2(5) = 107.6769$, $p < .001$), Recall@5 ($\chi^2(5) = 52.5064$, $p < .001$), MRR ($\chi^2(5) = 69.4665$, $p < .001$), and nDCG@5 ($\chi^2(5) = 37.6621$, $p < .001$). Holm-corrected pairwise analysis identified 30 significant comparisons among the 60 pairwise comparisons, with 19 large and 11 moderate rank-biserial effect sizes.

The Hybrid BM25–E5 configuration recorded the highest descriptive value across all five retrieval measures and was significantly different from every individual retriever for Precision@5, MRR, and nDCG@5. For Recall@5, it differed significantly from four of the five individual models; the comparison with BM25 was the exception ($p = 0.1163$). The hybrid configuration was consequently carried forward to the generation and abstention stages.

The RQ1 and RQ2 findings are not contradictory. RQ1 compared retrieval paradigms at the family level, whereas RQ2 examined individual configurations and their combination. The absence of a significant aggregate semantic advantage does not imply that semantic information is redundant. Rather, the hybrid results indicate that the lexical and semantic mechanisms can contribute different evidence even when neither family independently establishes a statistically significant advantage over the other.

The evidence-composition analysis provides direct support for this interpretation. Among the 675 chunks appearing in the Hybrid top-5 results, 46.96% were retrieved by both BM25 and E5, while 27.56% were surfaced only by BM25 and 25.48% only by E5. Thus, more than half of the final hybrid evidence consisted of chunks contributed exclusively by one of the two retrieval mechanisms. The RAW013 example further illustrates this complementarity: its five hybrid results contained two BM25-only chunks and three E5-only chunks, with no chunk shared by both components. Reciprocal Rank Fusion therefore did not merely reproduce one component's ranking; it combined partially distinct retrieval signals.

This result is consistent with the literature reviewed in the study, which describes hybrid retrieval as a mechanism for combining lexical precision with semantic recall and notes that retrieval performance can vary across domains (Thakur et al., 2021). It also provides a methodological explanation for why selecting a single embedding model solely on the expectation of semantic superiority may be insufficient for this legal corpus. The evidence suggests that retrieval architecture matters not only through the quality of an individual representation but also through how different retrieval signals are combined.

At the same time, the RQ2 result should not be interpreted as demonstrating that Reciprocal Rank Fusion is universally optimal for legal retrieval. The experiment evaluated one specific hybrid configuration—unweighted RRF with BM25 and E5—under a fixed corpus, chunking strategy, and retrieval depth. The observed advantage therefore establishes the empirical behavior of this configuration under the study conditions rather than a general superiority claim for all hybrid retrieval approaches. :chatgpt-content-referenceindex="2"

5.3 RQ3: Evidence-Grounded Answer Generation

RQ3 investigated whether NyayaRAG could generate high-quality evidence-grounded answers after the Hybrid BM25–E5 retriever was fixed as the retrieval component. The generation experiment evaluated 60 Answerable questions using five dimensions: Legal Correctness, Evidence Grounding/Faithfulness, Citation Accuracy, Answer Relevance, and Completeness. The resulting mean composite score was 0.8983, and 52 of the 60 answers (86.67%) reached or exceeded the predefined acceptability threshold of 0.75. A one-sided one-sample Wilcoxon signed-rank test produced $W = 1452.0$, $p = 0.000013$, with a rank-biserial effect size of $r = 0.5869$.

The dimension-level results provide additional context for the composite score. Evidence Grounding had a mean score of 1.8833, Answer Relevance had a mean of 1.8667, and Legal Correctness had a mean of 1.7500. Citation Accuracy was perfect among the 46 answers for which that dimension was applicable. Completeness had the lowest mean at 1.6500 and therefore represented the principal residual weakness identified in the evaluation. These results indicate that the main remaining quality limitation was not the general relevance or evidence grounding of responses, but whether the generated answer fully covered the information required by the question. :chatgpt-content-referenceindex="3"

The finding is consistent with the conceptual motivation for retrieval-augmented generation described in the literature. Lewis et al. (2020) established RAG as an approach for conditioning generation on retrieved evidence rather than relying exclusively on parametric knowledge. The present study extends that principle to the evaluated Indian Supreme Court corpus by imposing an evidence-constrained generation prompt that explicitly instructed the model not to introduce unsupported legal knowledge, case names, citations, or statutory provisions. The observed grounding and relevance scores are consistent with the expected role of evidence-constrained generation in reducing reliance on unsupported content.

The result also relates to the legal-RAG literature emphasizing the importance of retrieval quality for downstream answer quality. Pipitone and Houir Alam (2024) reported that retrieval quality strongly constrains legal answer quality. In NyayaRAG, the generation evaluation was deliberately performed only after the Hybrid BM25–E5 configuration had been selected through the retrieval experiment. This sequencing provides methodological separation between retrieval selection and downstream generation evaluation, although it does not establish a causal relationship between the hybrid retriever and the observed answer-quality score.

The RQ3 findings should therefore be understood as evidence for the performance of the complete frozen configuration rather than for Gemini or the prompt independently. The experiment used a single generation model, Google's `gemini-3.6-flash`, with temperature 0.0, top-p 1.0, a maximum output allowance of 1,200 tokens, five retrieved evidence blocks, and a fixed citation-constrained prompt. Alternative generation models, prompts, and decoding strategies were not evaluated. Consequently, the result demonstrates the behavior of this particular retrieval-generation configuration rather than establishing that the selected generation model or prompt is optimal.

The completeness finding is particularly important from a methodological perspective. A response can remain relevant and well grounded while failing to include all information required by the question. This distinction is important in legal research because retrieval correctness alone does not guarantee that the final response adequately represents the relevant legal material. The study's five-dimensional rubric therefore provides a broader interpretation of answer quality than retrieval metrics alone. This is consistent with the literature's emphasis on evaluating generated legal answers rather than treating retrieval performance as a complete proxy for system usefulness.

5.4 RQ4: Abstention Under Insufficient Evidence

RQ4 evaluated whether NyayaRAG could recognize insufficient evidence and abstain rather than produce an unsupported answer. Among the 60 Unanswerable questions, 59 were correctly abstained on, corresponding to a safe-abstention rate of 98.33%. The exact one-sided binomial test against the predefined 0.80 benchmark produced $p = 0.000025$, with a 95% confidence interval of [0.9234, 1.0000] and Cohen's $h = 0.6684$. The observed safe-abstention proportion therefore exceeded the prespecified benchmark under the study's statistical criterion.

This finding is particularly relevant to the reliability concerns identified in the legal-AI literature. The literature review notes that retrieval grounding reduces hallucination risk but does not eliminate it, while prior evaluations have documented unsupported or incorrect legal citations even in systems marketed as grounded. Research on abstention similarly treats the ability to withhold an answer when evidence is insufficient as an important reliability property (Wen et al., 2024; Nguyen et al., 2025; Richmond et al., 2023). The RQ4 result is therefore consistent with the study's broader design principle that legal QA systems should distinguish between evidence-supported and evidence-insufficient queries rather than treating every question as answerable. :chatgpt-content-referenceindex="4"

The safety result, however, needs to be interpreted together with the answerability results. The combined 120-question confusion matrix contained 46 true positives, 14 false negatives, one false positive, and 59 true negatives. The resulting precision was 97.87%, recall was 76.67%, and F1-score was 85.98%. The more frequent observed classification error was therefore over-abstention: 14 Answerable questions were classified as abstentions. In contrast, only one of the 60 Unanswerable questions received an unsupported answer, corresponding to an unsafe-answer rate of 1.67%.

This pattern illustrates a safety–coverage trade-off within the evaluated configuration. A system that abstains conservatively can reduce unsupported answers while also

declining to answer some questions for which sufficient evidence exists. The study's results therefore do not support interpreting the high abstention rate in isolation as evidence of complete reliability. Instead, safe abstention and answer coverage must be considered jointly.

The report also identifies three answerability edge cases—RAW107, RAW116, and RAW120—where the retrieved evidence addressed the general legal area but did not establish the specific facts required by the question. These cases illustrate that legal answerability is not determined solely by whether retrieved material is topically related to a query. Evidence must also contain the factual or legal information necessary to support the requested conclusion. This distinction is central to the practical interpretation of an abstention-capable legal RAG system.

5.5 Methodological Interpretation

Taken together, the four research questions indicate that retrieval and generation should be considered as connected stages rather than independent components. The retrieval experiment showed that semantic retrieval did not provide a statistically significant family-level advantage over lexical retrieval, while the six-way comparison showed that a hybrid configuration combining BM25 and E5 produced statistically differentiated retrieval performance. The downstream experiments then evaluated this selected retrieval configuration under a fixed evidence-grounded generation protocol.

The methodological design also benefited from controlling several factors across retrieval configurations. The same 2024 corpus, paragraph-aware chunking strategy, benchmark, retrieval depth, and evaluation framework were used across the retrieval comparison, allowing retrieval configuration to remain the principal experimental variable. The final chunking configuration, `chunk_384_40`, contained 30,521 chunks and was selected because it provided a balance among granularity, coherence, and efficiency. The common corpus and fixed experimental conditions strengthen the internal comparability of the retrieval results, although they do not remove the broader limitations of the benchmark or corpus scope. :chatgpt-content-referenceindex="5"

The statistical design also provides an important interpretive boundary. Nonparametric tests were selected for repeated-measures comparisons, and RQ2 employed Holm correction for the post-hoc pairwise comparisons. However, the three RQ1 Wilcoxon tests were not family-wise corrected, whereas RQ2 applied Holm correction. Kendall's W, originally planned as an additional RQ2 effect-size measure, was not computed in the final analysis. These deviations were disclosed in the report and should be retained when interpreting the statistical evidence.

5.6 Practical Implications

Within the evaluated scope, the results indicate several practical implications for legal research-support systems. First, retrieval architecture is an important design consideration. The findings do not support relying on semantic retrieval solely because legal queries may contain conceptual variation. Instead, the benchmark showed measurable value in combining lexical and semantic retrieval signals through BM25–E5 RRF.

Second, evidence traceability remains central to the system design. NyayaRAG supplies retrieved evidence alongside generated answers, allowing professional users to inspect the underlying judgment material rather than treating the generated response as an independent legal conclusion. This is consistent with the legal-research requirement for source traceability emphasized throughout the study.

Third, abstention provides a mechanism for handling evidence insufficiency. The 98.33% safe-abstention rate on the evaluated Unanswerable benchmark demonstrates that the system frequently withheld an answer when the evidence was insufficient. However, the 14 observed over-abstentions demonstrate that such conservatism can reduce answer coverage. For professional users, the appropriate interpretation is therefore that the system can function as a research-support layer while retaining human review of retrieved authorities and generated responses.

The intended user population also defines the scope of these implications. The benchmark was constructed around judicial officers and the report identifies judicial officers, advocates, and legal researchers as the intended professional users. The study did not evaluate citizen or lay-user participants and therefore does not establish suitability for direct citizen-facing legal advice. Similarly, no wall-clock benchmarking was performed, so claims about time savings remain structural implications rather than measured performance improvements.

5.7 Unexpected and Null Findings

The most important unexpected result was the absence of a statistically significant semantic-retrieval advantage in RQ1. The literature review had established several reasons to expect semantic retrieval to outperform lexical approaches, particularly in the presence of vocabulary mismatch. However, the benchmark did not reproduce that expected pattern. The result highlights the importance of evaluating retrieval strategies empirically within the target legal domain rather than assuming that improvements observed in general-domain or adjacent legal tasks will transfer directly.

The contrast between RQ1 and RQ2 is particularly informative. Semantic retrieval did not outperform lexical retrieval as a family, yet combining BM25 and E5 yielded the strongest retrieval configuration among those evaluated. This indicates that the absence of a family-level semantic advantage does not imply that semantic representations have no useful contribution. Instead, the evidence suggests that their contribution can become more apparent when combined with lexical evidence.

A second important negative finding concerns coverage. Although the system achieved a high safe-abstention rate, 14 of the 60 Answerable questions were classified as abstentions. This indicates that conservative abstention can produce a meaningful coverage cost. Completeness was also the lowest-scoring dimension of the RQ3 answer-quality rubric, reinforcing that producing a grounded response does not necessarily guarantee complete coverage of all information requested.

5.8 Reliability and Interpretation Boundaries

The findings provide strong internal evidence for the specific evaluation conditions but should not be generalized beyond them without additional validation. The corpus

consisted exclusively of 782 English-language Supreme Court of India judgments from 2024. The results therefore do not establish performance for other years, High Courts, subordinate courts, tribunals, or non-English judgments.

The benchmark was derived from questions contributed by 29 judicial officers. Although this provides professional grounding for the evaluation, the respondent pool was modest and non-random. In addition, the annotation process did not include a formally calculated inter-rater reliability statistic or independent second-annotator cross-check. These factors limit the extent to which the relevance and answerability labels can be treated as universally reproducible judgments.

The retrieval and generation configurations also impose boundaries. Retrieval was fixed at $k = 5$, so relevant material outside the five retrieved chunks was unavailable to the generator. The embedding models were not fine-tuned on the corpus or benchmark, and the study did not systematically compare alternative chunking configurations during the final evaluation. RQ3 and RQ4 evaluated one generation model, one prompt, and one decoding configuration. Accordingly, the observed results characterize the evaluated pipeline rather than an optimized search over all possible configurations.

There are also operational limitations. The deployed Gradio interface was a proof-of-concept rather than a production deployment. Security, load, privacy, institutional access control, scalability, and latency were not formally evaluated. The system relied on the Google Gemini API, introducing provider availability, connectivity, and cost considerations that were outside the experimental evaluation. Furthermore, the system did not model legal-authority hierarchy, bench strength, subsequent treatment, or overruling. A citation retrieved from the 2024 corpus therefore does not by itself establish the current precedential status of an authority.

Finally, the automated behavioral classification used in the combined confusion matrix relied on substring matching. An earlier pre-screen classified the RQ3 set as 50 Answered and 10 Abstained, whereas the final classification yielded 46 Answered and 14 Abstained. Because these classifications were not independently re-verified manually, the precise false-negative count should be interpreted with this documented caveat. This does not alter the reported RQ4 safe-abstention result of 59 correct abstentions among 60 Unanswerable questions, but it limits the precision with which the combined confusion-matrix error distribution should be interpreted. :chatgpt-content-referenceindex="6"

5.9 Implications for Future Research

The findings identify several directions for extending the evaluation. First, future studies can compare multiple generation models, prompts, and decoding strategies under identical frozen-evidence conditions to determine whether the observed RQ3 performance is robust to generation-model choice. Automated citation verification against corpus metadata could also reduce dependence on manual citation assessment.

Second, the over-abstention pattern suggests investigating calibrated evidence-sufficiency mechanisms that can improve answer coverage without increasing unsafe-answer rates. Such mechanisms could expose interpretable evidence-sufficiency indicators rather than relying solely on an opaque generation decision.

Third, a larger and more heterogeneous benchmark with multiple independent annotators and formally calculated inter-rater reliability would strengthen the evaluation of relevance and answerability. Extending the corpus beyond the 2024 English-language Supreme Court snapshot would also permit examination of temporal and institutional generalizability.

Finally, future versions could incorporate legal-authority structure, including citation networks, bench strength, and subsequent treatment such as overruling or distinguishing. The report also identifies query rewriting and clarification of ambiguous questions as potential extensions, alongside a broader judicial research environment for issue-based precedent discovery and authority comparison. These directions follow directly from the limitations identified in the current evaluation rather than representing capabilities already demonstrated by the present system. :chatgpt-content-referenceindex="7"

6 Implementation and User Benefit

6.1 Deployment Approach

The originally proposed deployment architecture consisted of a local, privacy-oriented stack based on OpenWebUI, Docker, ChromaDB, and Ollama. However, this configuration was not used in the final implementation. The demonstrated system was instead deployed as a conversational Gradio application using Gradio version 6.20.0. The deployed implementation therefore represents a proof-of-concept demonstration of the validated NyayaRAG pipeline rather than the originally proposed local deployment architecture.

The deployed application carries forward the retrieval and generation configuration validated during the evaluation without conducting a separate re-evaluation at deployment time. The operational workflow follows the sequence *user query → hybrid retrieval → top-5 evidence → grounded generation → citation/evidence display → answer or abstention*. The retrieval component uses the Hybrid BM25–E5 configuration selected during RQ2 and subsequently retained for the RQ3/RQ4 generation evaluation.

An additional data-completeness issue was addressed during deployment. The retrieval chunk corpus did not originally contain a dedicated formal-citation field. An exact mapping was therefore established between `judgment_id` and the corresponding judgment-level citation across all 782 unique judgments. This mapping was subsequently used to backfill formal citations, such as "[2024] 2 S.C.R. 943", into the deployment layer. The interface presents deduplicated authorities under a *Sources* heading, using the format *Case Title, Formal Citation*. This citation enhancement was introduced specifically for deployment and therefore postdates the RQ3/RQ4 evaluation, where citation accuracy was assessed using case-title/judgment-ID style references.

The final interface supports continuous multi-turn interaction under the title *NyayaRAG — Conversational Supreme Court Legal Research*. Each user turn initiates a new Hybrid BM25–E5 retrieval operation; retrieval results are not cached between turns. When conversational history is available, a lightweight follow-up-aware query

construction mechanism incorporates the immediately preceding user turn to support follow-up queries. Up to five previous conversational turns are supplied to the generation model for reference resolution, while the substantive legal content of each response is restricted to evidence freshly retrieved for the current turn.

Generation is performed through Google's Gemini API using `gemini-3.6-flash`, with temperature set to 0.0 and top-p set to 1.0. The configuration corresponds to the frozen RQ3/RQ4 generation setup, except that the maximum output allowance was increased from 1,200 to 2,000 tokens for the live demonstration to reduce truncation in longer conversational responses. The fixed abstention response, "The retrieved evidence is insufficient to provide a reliable answer," is retained unchanged from the evaluation configuration.

The application was launched using `demo.queue().launch(share=True, debug=False)`, which generated a temporary public Gradio share link with a validity period of up to one week. This configuration was suitable for proof-of-concept demonstration but was not intended as sustained institutional hosting. No formal latency or hardware benchmarking was performed, and the deployed interface was not subjected to security, load, privacy, or institutional access-control testing. Consequently, the implementation should be regarded as a functional feasibility demonstration rather than a production deployment.

6.2 System Integration

NyayaRAG is implemented as a legal research-support layer rather than as an autonomous case-management or decision-making system. Its principal integration point is therefore within existing legal research workflows, where retrieved precedents and generated responses can supplement rather than replace established judgment databases, case-management systems, and professional legal review.

For judicial officers and their research staff, the system can function as a precedent discovery aid. A legal research question can be submitted in natural language, after which the system retrieves relevant portions of the 2024 Indian Supreme Court corpus and produces a citation-anchored response. The resulting authorities remain available for independent verification against the complete judgment before being relied upon. For advocates, the same query-to-cited-answer workflow can support case preparation and the initial stages of brief development. The conversational design also permits follow-up questions within a single research session, which is applicable to law students, researchers, and legal-aid organizations identified as secondary user groups in the project.

The current implementation remains a standalone Gradio application rather than an embedded component within an existing legal information platform. Direct integration with e-Court systems, judgment databases, or case-management software would therefore require additional engineering. Possible integration mechanisms identified by the project include exposing the retrieval-generation pipeline through an API or embedding the interface within an institutional portal with suitable authentication.

The architecture nevertheless provides a basis for such integration because the retrieval and generation components are separated from the user-interface layer. Furthermore, the citation-backfill mechanism establishes a direct relationship between

retrieved chunks and formal judgment metadata, allowing generated results to be resolved against the corresponding source judgments.

6.3 User Interaction

The user-facing interaction is based on natural-language legal questions and does not require Boolean operators, specialized retrieval syntax, or manually constructed keyword queries. Each submitted question follows the same processing path: the query, optionally incorporating the immediately preceding conversational turn, is passed to the Hybrid BM25–E5 retrieval layer, which returns the five highest-ranked evidence chunks. These chunks, together with their case titles, judgment identifiers, and formal citation information, are supplied to Gemini under the grounded-generation prompt.

The system produces one of two output modes. When the retrieved evidence is sufficient, NyayaRAG generates a structured answer that addresses the question and identifies the supporting legal principles or holdings. Substantive claims are constrained to the retrieved evidence, and the response is accompanied by relevant case references. A deduplicated list of supporting authorities is subsequently displayed under a *Sources* heading.

When the retrieved evidence is insufficient, the system does not generate a substantive legal response. Instead, it returns the fixed abstention message: *"The retrieved evidence is insufficient to provide a reliable answer."* The generation instructions explicitly prohibit the model from reconstructing, supplementing, or embellishing legal content using information outside the retrieved evidence.

Internal retrieval information, including BM25 and E5 scores, ranks, Reciprocal Rank Fusion values, and chunk identifiers, is not exposed to the user. The interface therefore emphasizes the answer and its supporting authorities rather than the internal retrieval diagnostics.

The deployed interface provides four preloaded example questions to assist first-time users. These concern amendment of pleadings under Order VI Rule 17 of the CPC, summoning an additional accused under Section 319 of the CrPC, safeguards concerning arrest and police remand, and principles governing specific performance. The interface was implemented using a Gradio `Blocks` container together with a `ChatInterface` component, providing continuous multi-turn interaction.

6.4 Benefits to Users

The practical benefits of NyayaRAG can be separated into demonstrated benefits, intended operational benefits, and potential future benefits. This distinction is important because the project evaluated retrieval, answer quality, and abstention behavior, but did not conduct a formal user-productivity or deployment-scale study.

6.4.1 Demonstrated Benefits

Traceability.

Traceability was directly supported by the system's citation-constrained generation design. In the RQ3 evaluation, the citation-accuracy dimension received a score of 2.00, corresponding to perfect citation accuracy wherever a citation was applicable,

covering 46 of the 60 evaluated answers. The deployed interface further appends a deduplicated list of formal citations under the *Sources* heading, allowing users to trace substantive claims back to the retrieved judgment material.

Evidence-based abstention.

The system also demonstrated a measurable safety property under deliberately unanswerable conditions. In the 60-question RQ4 benchmark, NyayaRAG correctly abstained on 59 questions, corresponding to a safe-abstention rate of 98.33%. An unsupported answer occurred in 1 of the 60 cases, or 1.67%. Across the combined 120-question RQ3/RQ4 confusion matrix, the reported precision was 97.87%.

This safety behavior must nevertheless be considered together with the observed coverage limitation: 14 of the 60 answerable questions were incorrectly abstained upon. Thus, abstention reduces the risk of unsupported answers but can also prevent answers when relevant evidence exists.

Structural search efficiency.

The natural-language interaction reduces the need for repeated query reformulation associated with conventional keyword or Boolean retrieval when terminology differs between a user's question and the relevant judgment. The benefit demonstrated by the architecture is therefore structural: one natural-language query initiates a retrieval and generation cycle. However, the study did not perform formal wall-clock or query-latency benchmarking. Consequently, no quantitative claim about actual time saved, aggregate productivity, or backlog reduction is made.

6.4.2 Intended Operational Benefits

For judicial officers and legal researchers, the intended benefit is to provide a more direct mechanism for locating relevant Supreme Court precedent and examining supporting legal material. For advocates, the same mechanism is intended to support case preparation and preliminary legal research. Judicial academies may also use the conversational interface as a training aid for exploring Supreme Court judgments and legal principles.

These uses remain research-support applications. The system is not intended to substitute for professional legal judgment, and users remain responsible for verifying the retrieved authorities and assessing their relevance, interpretation, precedential status, and validity.

6.4.3 Potential Future Benefits

Future institutional deployment could potentially extend the system to larger and more diverse judgment collections and connect it directly with existing legal research or case-management platforms. Such extensions could provide broader access to precedent and more integrated research workflows. These benefits remain prospective because the present study was restricted to 782 English-language Supreme Court judgments from 2024 and the demonstrated interface was not subjected to institutional deployment testing.

6.5 Example Use Case

A representative live interaction demonstrates the intended end-to-end behavior of NyayaRAG. The user submitted the natural-language question:

"What principles govern amendment of pleadings under Order VI Rule 17 CPC?"

The Hybrid BM25–E5 retriever returned five evidence chunks. Two originated from *Basavaraj v. Indira and Others*, [2024] 2 S.C.R. 935, one from *Mallavva and Anr. v. Kalsammanavara Kalamma (Since Dead)*, [2024] 12 S.C.R. 1884, and two from *Sheikh Noorul Hassan v. Nahakpam Indrajit Singh & Ors.*, [2024] 6 S.C.R. 53. The reported Reciprocal Rank Fusion scores were approximately 0.0296–0.0306.

Gemini subsequently generated a principle-oriented response concerning amendment of pleadings under Order VI Rule 17 CPC. The response addressed several issues supported by the retrieved authorities, including that amendment is not an unrestricted matter of right, the avoidance of multiplicity of litigation, restrictions on changing the fundamental character of a suit, the additional requirements applicable after commencement of trial, considerations concerning bona fides and prejudice, and the powers of the first appellate court under Section 107(2) CPC.

The interface displayed the supporting authorities separately as an "Authorities Relied Upon" list. This interaction demonstrates the complete operational sequence: a natural-language question is submitted, relevant evidence is retrieved, the evidence is synthesized into a structured response, and the supporting judgments are exposed for independent verification.

6.6 Recommendations for Adoption and Use

The project findings and deployment experience support several recommendations for responsible use of NyayaRAG as an Indian Supreme Court legal-research aid.

First, the system should be positioned as a research assistant rather than a decision-making system. Its appropriate role is to assist judges, lawyers, and research staff in locating and examining relevant judgments; the final legal conclusion should remain with the qualified professional using the system.

Second, legal professionals should verify cited authorities against the underlying judgments before relying on generated content. Although the evaluated citation accuracy was perfect wherever applicable, verification remains necessary because retrieval and generation errors are still possible.

Third, institutions considering adoption should begin with a limited deployment involving a small user group and a controlled collection of reliable judgments. Gradual expansion would permit assessment of system behavior, evidence coverage, and the trade-off between safe abstention and over-abstention under operational conditions.

Fourth, abstention should be treated as a central safety requirement. The ability to decline to provide an answer when available evidence is inadequate is an explicit design property of NyayaRAG and was evaluated through RQ4.

Finally, institutional users should prioritize systems that expose the evidentiary basis of generated responses and permit verification against authoritative legal sources.

For NyayaRAG, the citation-anchored response and formal "Sources" section provide this traceability mechanism. The principal operational value is therefore better characterized as improved access to relevant legal information and structural research efficiency rather than as a formally measured reduction in research time.

7 Limitations

Although the evaluation provides controlled evidence for the four research questions, several limitations constrain the interpretation, generalizability, and operational readiness of the findings. These limitations concern the corpus and benchmark, retrieval and generation configuration, statistical analysis, deployment environment, and the legal and ethical scope of the system.

7.1 Dataset and Scope Limitations

The evaluation corpus is restricted to 782 English-language judgments of the Supreme Court of India from the year 2024. Consequently, the reported findings should be interpreted specifically within this temporal, institutional, and linguistic scope. They do not establish equivalent performance for judgments from other years, High Courts, subordinate courts, tribunals, or non-English legal material. The single-year corpus was an intentional scoping decision for establishing and validating the retrieval-augmented generation pipeline, but it limits the extent to which the observed results can be generalized to the broader Indian legal domain.

A related documentation limitation concerns the processed corpus artifacts. The exact composition of the 19-column `metadata_2024_processed.parquet` file beyond the seven explicitly identified derived fields was not fully itemized in the source notebook, while two archived snapshots contained 27 and 32 columns without corresponding itemized manifests. In addition, the initial duplicate-`case_id` check reported 37 apparent duplicates, although this count was inflated by blank-value matching; after excluding these spurious matches, only one genuine duplicate pair was identified, and no record was removed. This documentation issue does not alter the final corpus, but it should be considered when reproducing the data-preparation process.

7.2 Benchmark and Annotation Limitations

The evaluation benchmark was derived from questions contributed by 29 judicial officers. This design intentionally prioritizes professional legal-research needs over crowd-sourced or synthetic questioning; nevertheless, the respondent pool remains modest and non-random. The resulting benchmark therefore provides evidence for the professional research context represented in the collected questions rather than a statistically representative sample of all possible Indian legal queries.

The relevance and answerability annotation process was conducted using structured rubrics, but the final workflow employed a single/limited-annotator protocol. No formal inter-rater reliability statistic, such as Cohen's kappa, was calculated, and completed labels were not independently cross-checked by a second annotator. Internal validation checks assessed consistency within the annotation workflow but were

not designed to quantify agreement between independent annotators. Accordingly, the reported evaluation does not establish inter-annotator agreement for the benchmark labels.

7.3 Retrieval and Generation Configuration Limitations

The retrieval stage used a fixed retrieval depth of $k = 5$ throughout the evaluation. Evidence ranked below the fifth position was therefore unavailable to the generation component, meaning that the downstream generation results characterize performance under this particular evidence budget rather than under unrestricted retrieval depth.

The chunking strategy was likewise fixed to a single selected configuration, and no systematic comparison of alternative chunking strategies was incorporated into the final retrieval evaluation. In addition, none of the evaluated embedding models was fine-tuned using the study corpus or benchmark. These choices limit the extent to which the reported retrieval performance can be interpreted as representative of optimized domain-adapted retrieval systems.

The generation and abstention evaluations were also based on a single configuration: `gemini-3.6-flash`, one prompt, and one decoding setting. Multiple language models, prompts, or decoding strategies were not compared under otherwise identical conditions. The resulting RQ3 and RQ4 measurements therefore characterize the effectiveness of the evaluated configuration rather than establishing that it is optimal. Sensitivity to model choice, prompt formulation, or decoding parameters was not directly evaluated.

The system further relies on a single hosted third-party API, Google Gemini. Consequently, connectivity, latency, API cost, and provider availability represent operational dependencies that were not evaluated experimentally in this study.

7.4 Statistical and Methodological Limitations

The statistical analyses for RQ1 and RQ2 were not fully symmetrical in their treatment of multiple comparisons. The three Wilcoxon signed-rank tests conducted for RQ1 were not subjected to family-wise error correction, whereas the six-way RQ2 comparison incorporated Holm correction for the post-hoc pairwise tests. This difference in inferential treatment should be considered when interpreting the RQ1 results.

In addition, the non-significant RQ1 result should not be interpreted as evidence that lexical and semantic retrieval are statistically equivalent. Rather, it indicates that the analysis did not provide sufficient evidence to support the pre-specified hypothesis that semantic retrieval significantly outperforms lexical retrieval.

A further methodological deviation concerns RQ2. Kendall's W was specified in the original analysis plan as an additional effect-size measure but was not computed in the final analysis. The final RQ2 reporting therefore relies on the Friedman statistic, degrees of freedom, p-value, Holm-corrected pairwise Wilcoxon comparisons, and rank-biserial effect sizes. This deviation was explicitly disclosed and does not represent a post-hoc omission.

7.5 Deployment and Operational Limitations

The deployed Gradio interface represents a proof-of-concept implementation rather than a production deployment. It was not subjected to security, load, privacy, institutional access-control, or broader operational testing. Similarly, scalability and latency were not benchmarked. The temporary public share link, which is available for up to one week, should therefore not be regarded as a substitute for institutionally hosted deployment.

The absence of formal wall-clock benchmarking also limits claims concerning operational efficiency. Although the system is intended to reduce the effort required to locate relevant judgments and provide a useful starting point for professional legal research, the study did not formally measure time savings or end-to-end search latency.

7.6 Legal and Ethical Limitations

NyayaRAG is intended as a legal research assistance system rather than an autonomous decision-making system. Responsibility for assessing the relevance, interpretation, precedential status, and current validity of retrieved authorities remains with the qualified user. The evaluation therefore does not support the use of the system as a substitute for professional legal or judicial judgment.

The system also does not explicitly model the hierarchy or continuing status of legal authorities. In particular, bench strength, subsequent treatment, and events such as overruling are not modeled. Consequently, the presence of a cited passage in the retrieved corpus does not by itself establish that the passage represents the current state of the law beyond the 2024 corpus snapshot.

Despite the strong aggregate RQ3 and RQ4 results, residual unsafe behavior was observed: the final evaluation recorded a 1.67% unsafe-answer rate. Thus, evidence-grounded generation and abstention reduce the risk of unsupported responses within the evaluated setting but do not eliminate the possibility of an incorrect answer when evidence is insufficient or misapplied.

7.7 Pilot-Stage Evaluation Limitation

An additional limitation arose during the preliminary abstention pilot. For Question ID RAW015, the ground-truth answerability label was changed from "answerable" to "insufficient_evidence" after the model had already abstained on the question. Because the relabeling occurred after observing model behavior, the pilot abstention metric consequently registered as perfect. This creates a risk of evaluation-set contamination and means that the pilot abstention statistic should not be interpreted as a confirmatory result.

The issue was confined to the pilot-stage validation process. In the final RQ4 evaluation, comprising $N = 60$ unanswerable questions, ground-truth labels were fixed before generation and were not reopened after model outputs were observed. The pilot results therefore remain applicable to their intended role as pipeline-validation evidence, whereas the final RQ4 evaluation constitutes the hypothesis-tested abstention result.

7.8 Overall Scope of the Limitations

Taken together, these limitations primarily constrain the scope within which the findings should be interpreted rather than invalidate the controlled comparisons performed in the study. The corpus and benchmark restrictions limit generalization beyond English-language 2024 Supreme Court question answering and beyond the annotation procedure used. The fixed retrieval and generation settings mean that the reported metrics describe a particular pipeline configuration rather than an exhaustively optimized system. Finally, the proof-of-concept deployment, absence of operational benchmarking, and lack of legal-authority hierarchy modeling limit direct interpretation of the system as a production or institutionally certified legal technology.

Accordingly, the empirical findings should be understood as evidence obtained under the specified corpus, benchmark, annotation, retrieval, generation, and deployment conditions. Broader claims concerning other legal jurisdictions within India, other languages or time periods, alternative model configurations, multi-annotator evaluation, or production-scale operation require additional validation.

8 Future Work

Several directions follow naturally from the scope and limitations of the present study. The current evaluation establishes evidence for English-language Supreme Court judgments from 2024, but the corpus and experimental configuration remain intentionally bounded. Future research can therefore extend the empirical scope of NyayaRAG while preserving its evidence-grounded and abstention-oriented design.

8.1 Corpus Expansion and Multilingual Coverage

A first direction is to broaden the corpus beyond the 2024 Supreme Court collection. Future versions could incorporate a historical judgment range from 1950 to 2025 and extend coverage to High Court judgments through the eCourts API. Such expansion would allow the retrieval framework to be evaluated across a substantially wider temporal and institutional range rather than within a single-year Supreme Court snapshot. Hindi-language judgments and multilingual support could also be incorporated to examine whether the observed retrieval patterns extend beyond the English-language setting evaluated in the present study.

8.2 Retrieval and Methodological Improvements

The retrieval configuration can be investigated more extensively in subsequent work. The present system uses a fixed retrieval depth of $k = 5$, a single paragraph-aware chunking configuration, and embedding models that were not fine-tuned on the study corpus or benchmark. Future experiments could therefore compare alternative chunking approaches, including recursive, semantic, and judgment-structure-aware strategies. Fine-tuned or newer legal-domain embedding models could likewise be evaluated under otherwise comparable conditions. A reranking stage following the Hybrid BM25–E5 retrieval process could further be investigated to determine whether

additional evidence ordering improves the material presented to the generation component. As corpus size increases, migration from the current indexing configuration to a production-grade vector index would also be appropriate.

8.3 Generation and Citation Verification

The generation component provides another avenue for controlled experimentation. The current evaluation uses a single generation model, prompt, and decoding configuration. Future studies could compare multiple language models, prompt formulations, and decoding strategies while holding the retrieved evidence fixed. Such an evaluation would help separate the contribution of retrieval quality from generation-model and prompt sensitivity. Automated citation verification against corpus metadata could additionally be incorporated to strengthen the verification of citations produced by the generation stage.

8.4 Abstention and Evidence Sufficiency

The abstention mechanism could be extended through calibrated and tunable evidence-sufficiency thresholds. This would allow future experiments to examine the trade-off between answering more questions and maintaining a low unsafe-answer rate. The system could also expose interpretable indicators of evidence sufficiency to users rather than relying on an opaque internal decision. These extensions are particularly relevant because the current system demonstrates strong abstention performance while retaining a residual unsafe-answer rate.

8.5 Benchmark and Evaluation Expansion

Future evaluation should employ a larger and more heterogeneous benchmark together with multiple independent annotators. This would broaden the diversity of evaluation questions and permit formal measurement of inter-rater reliability. Such an evaluation would provide stronger evidence regarding the robustness of relevance and answerability labels than the current single- or limited-annotator protocol.

Further work could also incorporate the citation-network and authority characteristics that are currently absent from the retrieval framework. In particular, modeling bench strength, subsequent treatment, and overruling could provide information about the current precedential status of retrieved authorities. This would complement passage-level relevance by providing additional context concerning the legal authority and currency of retrieved material.

8.6 System Deployment and Human-Centered Evaluation

The current Gradio implementation is a proof-of-concept rather than a production deployment. Future development should therefore address security, load, privacy, scalability, latency, and institutional access control before operational deployment is considered. Offline or institutionally hosted language models could also be explored to reduce dependence on a single hosted third-party API.

Human-centered evaluation represents a further important extension. Structured evaluation with judicial officers, advocates, and legal researchers could assess the system in realistic research workflows and provide evidence beyond the controlled benchmark setting. Such evaluation would remain consistent with the intended role of NyayaRAG as a legal research assistance system rather than an autonomous decision-making system.

8.7 Conversational and Research-Environment Extensions

Beyond the improvements directly associated with the identified limitations, future work could improve conversational query understanding through query rewriting and clarification of ambiguous legal questions. In the longer term, NyayaRAG could evolve from a question-answering prototype into a broader judicial research environment supporting issue-based precedent discovery and comparison of legal authorities. These extensions would broaden the system's research-support capabilities while retaining the evidence-grounded orientation established in the present study.

9 Conclusion

This study addressed the problem of developing and empirically evaluating a retrieval-augmented generation framework for question answering over Indian Supreme Court judgments. The work focused on whether different lexical and semantic retrieval approaches could reliably identify relevant legal evidence, whether their combination could improve retrieval effectiveness, and whether a generation pipeline could produce evidence-grounded answers while appropriately abstaining when the available evidence was insufficient. The evaluation was conducted on the 2024 Supreme Court corpus and the benchmark, models, and configurations defined for the study.

The proposed NYAYaRAG framework combined corpus preprocessing and paragraph-aware chunking with six retrieval configurations: TF-IDF, BM25, E5, BGE, InLegalBERT, and a Hybrid BM25–E5 retriever using Reciprocal Rank Fusion (RRF). Retrieved evidence was subsequently supplied to the generation component, with the final system incorporating citation display and an abstention mechanism. The evaluation was organized around four research questions covering lexical–semantic retrieval differences, comparative retriever performance, evidence-grounded answer quality, and safe abstention.

The findings provide several distinct conclusions. For RQ1, the results did not provide statistically significant evidence that semantic retrieval models as a family outperformed lexical retrieval on Precision@5, MRR, or nDCG@5. The corresponding one-sided paired Wilcoxon signed-rank tests produced $p = 0.8176$, $p = 0.2868$, and $p = 0.5675$, respectively. Thus, the study did not reject the null hypothesis for the lexical–semantic family comparison. This result indicates that, within the evaluated corpus and experimental configuration, semantic retrieval could not be treated as uniformly superior to lexical retrieval.

RQ2 produced a different result at the level of individual retrieval configurations. The Hybrid BM25–E5 retriever recorded the highest value across all five evaluated retrieval metrics, and the Friedman tests were significant for all metrics ($p < .001$).

Holm-corrected pairwise comparisons further supported the observed differences. The findings therefore demonstrate that combining lexical and semantic retrieval through RRF produced a retrieval configuration with stronger measured performance than the individual models evaluated in this study. The complementary composition of the Hybrid evidence, in which BM25-only and E5-only chunks together accounted for more than half of the final retrieved evidence, further supports the interpretation that the two retrieval mechanisms contributed partially distinct evidence.

The generation experiments addressed RQ3. NYAYaRAG achieved a mean composite answer-quality score of 0.8983, with 86.67% of evaluated answers reaching the predefined threshold of 0.75. A one-sample Wilcoxon test yielded $W = 1452.0$, $p = 0.000013$, with an effect size of $r = 0.5869$, supporting the conclusion that the observed answer-quality distribution exceeded the specified benchmark. These results demonstrate the feasibility of generating answers grounded in the retrieved legal evidence under the evaluated configuration, while the citation-grounding results further support the system's source-traceability objective.

RQ4 examined whether the system could distinguish questions that should be answered from those for which the available evidence was insufficient. Among the 60 Unanswerable questions, 59 were correctly abstained on, corresponding to a safe-abstention rate of 98.33%, while one question produced an unsupported answer. The combined 120-question evaluation yielded an accuracy of 87.50%, precision of 97.87%, recall of 76.67%, F1-score of 85.98%, specificity/safe-abstention rate of 98.33%, and false-positive/unsafe-answer rate of 1.67%. The exact one-sided binomial test comparing the observed safe-abstention rate against the prespecified 0.80 benchmark produced $p = 0.000025$, with a 95% confidence interval of [0.9234, 1.0000] and Cohen's $h = 0.6684$. At the same time, 14 Answerable questions were incorrectly abstained on. Consequently, the principal classification error in the evaluated benchmark was over-abstention rather than unsafe answering, illustrating the safety–coverage trade-off inherent in the implemented mechanism.

Taken together, the results answer the four research questions in complementary ways. RQ1 did not establish a statistically significant advantage for semantic retrieval over lexical retrieval. RQ2 showed that combining lexical and semantic retrieval through Hybrid BM25–E5 produced significantly stronger retrieval performance than the individual configurations tested. RQ3 demonstrated that the resulting evidence-grounded generation pipeline exceeded its predefined answer-quality benchmark. RQ4 demonstrated statistically supported safe abstention under the evaluated benchmark, while also revealing a measurable coverage cost. These findings suggest that the principal methodological contribution of the study lies not in replacing lexical retrieval with semantic retrieval, but in integrating complementary retrieval mechanisms within an auditable RAG pipeline and evaluating both answer generation and abstention behavior explicitly.

The implemented system also provides practical significance within the demonstrated scope. The resulting pipeline connects user queries to Hybrid retrieval, Top-5 evidence selection, grounded generation, citation/evidence presentation, and either an answer or safe abstention. The deployment was demonstrated through a Gradio conversational interface, with formal citations reconstructed through the verified mapping

between `judgment_id` and the judgment-level citation field across the 782 unique judgments in the corpus. The resulting prototype therefore demonstrates how evidence retrieval, source traceability, grounded generation, and uncertainty-aware response behavior can be integrated into a single research-support workflow.

These conclusions should nevertheless be interpreted within the study's defined boundaries. The evaluation provides evidence for English-language question answering over the 2024 Indian Supreme Court corpus under the specific benchmark, annotation procedure, models, and fixed pipeline configuration used in the experiments. It does not establish generalizability to other years, other Indian courts, other languages, or multi-annotator evaluation. In addition, the system was demonstrated as a research-support prototype rather than a production or judicially certified system. The reported results therefore characterize the evaluated configuration rather than an exhaustively optimized legal question-answering system.

Overall, the study contributes an empirically evaluated framework for legal question answering that combines lexical and semantic retrieval, evidence-grounded generation, citation traceability, and explicit abstention. The results show that retrieval architecture, generation quality, and uncertainty handling should be evaluated jointly when assessing RAG systems for legal information access. Within the demonstrated scope, NYAYaRAG provides a reproducible experimental basis for studying these components over Indian Supreme Court judgments while making explicit the empirical and operational boundaries that must be addressed before broader deployment.

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